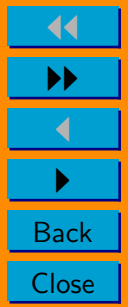


经 典 电 动 力 学

清华大学物理系

王 青



第二轮分组

A组: 陈静远, 王润丰, 金云韬, 刘力维, 高佳乐

B组: 朱逸天, 郑珺元, 罗景泓, 郭笑呈, 高鸿睿

C组: 王翟, 丁维轩, 卜少寒, 高辞寒, 张行楷

D组: 王丁尧, 杨丰瑞, 刘育辰, 行诗睿, 鲍康有

E组: 郑思哲, 吴桐, 伍衍璋, 马鸣远, 徐元璟

F组: 张子恒, 方子升, 张驰, 邵哲, 张博轩

G组: 柯贤睿, 诸葛宇轩, 王昱人, 方书衡, 赵鹏

H组: 张瑞涵, 吕潇冉, 郑玘锟, 傅越, 刘峻宇

I组: 施宇恒, 张行健, 黄飞禹, 黄意贤, 简铭

J组: 孙宁泽, 王博樊, 盛湛源, 朱昭宇, 熊宝博

K组: 刘宏润, 路致远, 黄浩, 汪泽元, 何楷然

L组: 姚安东, 陆张喆, 徐承恺, 徐骏阳, 邵美琪

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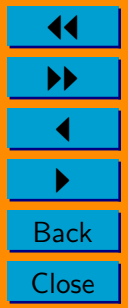
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问题1 （3月31日周一讨论）:6-6希格斯机制光子质量（42）； 6-7介质中光子质量（29）；

问题2 （4月2日周三讨论）:6-10量子化电磁波（35）； 6-23光子相关（24）；

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6-11四矢量（26）； 6-24复数问题（25）； 6-2非均匀介质（21）；

6-12边界条件（19）； 6-15方程的局限性（18）； 6-19偏振（18）；

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6-1介质（10）； 6-21全反射（9）； 6-13电场磁场（9）；



第3章 电磁波的传播第123节周一（3月31日）讨论题目1 6-6希格斯机制光子质量

1. 电磁波有纵波的形式吗？
2. 记得老师在费曼2的课上曾经提过超导体中的光子也有质量，这和普通的介质图像又有什么关系呢？
3. 在光子有质量的时候做的方程的修改为什么直接把质量就放到左边括号了？如果这样好像得到的还是质能方程，没什么新东西啊？
4. 介质中光子有质量或者介质非均匀都会导致方程非齐次，这两者有没有物理上的统一和相似之处？还有其他媒介子有质量的基本力在场描述下会不会也出现非齐次的波动方程？

第3章 电磁波的传播第123节周一（3月31日）讨论题目2 6-7介质中光子质量

1. 介质中光子有质量是否说明介质中光子可以“静止”？
2. 在哪些介质光子会有非零静止质量？
3. 如何从光子的角度理解电磁波在介质中速度小于光速？是否赋予了光子一个“质量”？那这个质量怎么来的？是与物质的相互作用吗？ 与希格斯机制有什么相同点或不同点？光子在介质中的静质量有实际含义吗？它是怎么出现的？
4. 光子的总能量E=hf，与介质无关，静质量这一部分多出来的能量哪来的，到哪去？为什么介质中光子会可能有质量？介质中光子有质量和介质 中光速小于c有联系吗？MOOC中提及在部分介质中光子会表现出质量，那么这会对应介质中电磁波的衰减（被吸收/复折射率）吗？这些性质间的联系应如何理解？ 如何推导慕课中提到的在介质中可能允许光子有质量的情况？介质参考系下光子有质量对应的物理模型是什么？
5. 介质中光子有质量或者介质非均匀都会导致方程非齐次，这两者有没有物理上的统一和相似之处？
6. 还有其他媒介子有质量的基本力在场描述下会不会也出现非齐次的波动方程？实际上，电磁波在晶体介质中由于晶格振动而受到阻力， 能不能说这样的阻力使得介质中光子看上去有了质量？
7. 如何写出真空中光子有质量情形的麦克斯韦方程组
8. 为什么如果光子有质量，真空波动方程就要加质量项修改?虽然这样修改后使方程能成立了，但和麦克斯韦方程组推出来的方程不一样， 难道意味着和麦克斯韦方程组矛盾吗？或者说质量项可以从介质的那几项中推出来？



第3章 电磁波的传播第123节周三（4月2日）讨论题目1 6-10量子化电磁波

1. 用量子力学中的算符改写电磁波的波动方程，是否意味着将电磁波看成某种粒子？
2. 电磁波中的守恒量对应什么对称性？
3. 量子中波动方程对应着质壳条件，但为什么非量子时二者的表达式不同？在非量子时这两个式子等价吗？
4. 麦克斯韦方程从量子角度看，E和B不用做修改吗？量子对应那个地方有啥意义吗，感觉就都是波的形式啊。为什么波动方程中的偏导算符可以直接和 量子力学中的动量算符和能量算符对应起来？
5. 那个从量子的观点看光子质量为零，为什么可以这样去看待问题，这种看待问题的方式是否是对电动力学有意义的？
6. 电磁场可以看作波函数吗？如果不能的话，为什么可以直接和量子的算符进行对应？

第3章 电磁波的传播第123节周三（4月2日）讨论题目2 6-23光子相关

1. 经典意义上的左右旋是如何过渡到光子自旋的认识的？
2. 自然光无固定相位差对应解释是怎样的呢？光子自旋处在非本征态？光子非零静质量？
3. 光子是自旋为1的粒子，为什么描述形式上却像一个自旋1/2的粒子？
4. 导体中电磁波传播的方向在电场方向有分量，是不是意味着此时光子的质量不为零？PPT里说弥散能量定义为不包含标准模型物质 粒子的能量构成形式，但是光子不也是标准模型粒子吗？
5. （非常无聊地撕定义）PPT说弥散能量包括”辐射”，那除了辐射还有什么？
6. 在讨论光子质量的过程中，量子与宏观的电磁场波动方程之间的对应是否是严格的？
7. 光子如何与介质相互作用，为什么光进入介质会减速？
8. 粒子运动的速度超过了介质中的光子会不会出现一些奇异的现象？



电磁波的传播







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麦克斯韦方程组的时间初值及时间演化

$$\nabla\cdot\vec{D}=\rho_f \qquad \nabla\times\vec{E}=-\frac{\partial\vec{B}}{\partial t} \qquad \vec{D}=\epsilon\vec{E}$$

$$\nabla\cdot\vec{B}=0 \qquad \nabla\times\vec{H}=\vec{j}_c+\frac{\partial\vec{D}}{\partial t} \qquad \vec{B}=\mu\vec{H}$$

散度方程　　旋度方程　　电磁性质方程

- 若在t_0时刻已知$\vec{E}(\vec{r},t_0)$，$\vec{B}(\vec{r},t_0)$；电磁性质方程$\Rightarrow$ $\vec{D}(\vec{r},t_0)$，$\vec{H}(\vec{r},t_0)$
- 旋度方程：时间演化$\Rightarrow\dot{\vec{D}}(\vec{r},t_0)$，$\dot{\vec{B}}(\vec{r},t_0)$； 电磁性质方程$\Rightarrow\dot{\vec{E}}(\vec{r},t_0)$，$\dot{\vec{H}}(\vec{r},t_0)$

  更自然的初始条件：已知$\dot{\vec{E}}(\vec{r},t_0)$，$\dot{\vec{B}}(\vec{r},t_0)$；电磁性质方程$\Rightarrow\dot{\vec{D}}(\vec{r},t_0)$，$\dot{\vec{H}}(\vec{r},t_0)$；散度旋度方程确定初始场强
- 就知道$t_0+\Delta t$时刻的场强：

$$\begin{aligned}\vec{E}(\vec{r},t_0+\Delta t)&=\vec{E}(\vec{r},t_0)+\dot{\vec{E}}(\vec{r},t_0)\Delta t\\ \vec{B}(\vec{r},t_0+\Delta t)&=\vec{B}(\vec{r},t_0)+\dot{\vec{B}}(\vec{r},t_0)\Delta t\\ \vec{D}(\vec{r},t_0+\Delta t)&=\vec{D}(\vec{r},t_0)+\dot{\vec{D}}(\vec{r},t_0)\Delta t\\ \vec{H}(\vec{r},t_0+\Delta t)&=\vec{H}(\vec{r},t_0)+\dot{\vec{H}}(\vec{r},t_0)\Delta t\end{aligned}$$

  循环往复，可知所有场强
- 散度方程：初始场强必须满足初始条件的约束；后继场强依赖旋度方程仍旧满足，

$$\begin{aligned}\nabla\cdot\Delta\vec{D}(\vec{r},t_0) &= \nabla\cdot\dot{\vec{D}}(\vec{r},t_0)\Delta t=\nabla\cdot\left[\nabla\times\vec{H}(\vec{r},t_0)-\vec{j}_c(\vec{r},t_0)\right]\Delta t=-\nabla\cdot\vec{j}_c(\vec{r},t_0)\Delta t\\ &= \dot{\rho}_f(\vec{r},t_0)\Delta t=\Delta\rho_f(\vec{r},t_0)\\ \nabla\cdot\Delta\vec{B}(\vec{r},t_0) &= \nabla\cdot\dot{\vec{B}}(\vec{r},t_0)\Delta t=\nabla\cdot\left[\nabla\times\vec{E}(\vec{r},t_0)\right]\Delta t=0\end{aligned}$$

从旋度方程可推出散度方程整体不随时间变化，散度方程的时间演化性质是不独立的！

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$$\vec{B}=\mu\vec{H} \qquad \vec{D}=\epsilon\vec{E}$$

理想绝缘介质中$\gamma=0$,考虑自由电荷和传导电流为零的情况,

$$\nabla\times\vec{E}=-\frac{\partial\vec{B}}{\partial t} \quad \frac{1}{\mu}\nabla\times\vec{B}+\nabla\frac{1}{\mu}\times\vec{B}=\epsilon\frac{\partial\vec{E}}{\partial t} \quad \epsilon\nabla\cdot\vec{E}=-\nabla\epsilon\cdot\vec{E} \quad \nabla\cdot\vec{B}=0$$

$$\nabla\times\vec{B}-(\nabla\ln\mu)\times\vec{B}=\mu\epsilon\frac{\partial\vec{E}}{\partial t} \qquad \nabla\cdot\vec{E}=-(\nabla\ln\epsilon)\cdot\vec{E}$$

$$\nabla^2\vec{E}=-\nabla[\nabla\cdot\vec{E}+(\nabla\ln\epsilon)\cdot\vec{E}]+\nabla^2\vec{E}=-\nabla\times(\nabla\times\vec{E})-\nabla[(\nabla\ln\epsilon)\cdot\vec{E}]$$

$$=\nabla\times\frac{\partial\vec{B}}{\partial t}-\nabla[(\nabla\ln\epsilon)\cdot\vec{E}]=\mu\epsilon\frac{\partial^2}{\partial t^2}\vec{E}+(\nabla\ln\mu)\times\frac{\partial\vec{B}}{\partial t}-\nabla[(\nabla\ln\epsilon)\cdot\vec{E}]$$

$$=\mu\epsilon\frac{\partial^2}{\partial t^2}\vec{E}-(\nabla\ln\mu)\times(\nabla\times\vec{E})-\nabla[(\nabla\ln\epsilon)\cdot\vec{E}]$$

$$=\mu\epsilon\frac{\partial^2}{\partial t^2}\vec{E}+(\nabla\ln\mu)\cdot\nabla\vec{E}-(\nabla\vec{E})\cdot(\nabla\ln\mu)-\nabla[(\nabla\ln\epsilon)\cdot\vec{E}]$$

$$\nabla^2\vec{E}-\frac{1}{v^2}\frac{\partial^2\vec{E}}{\partial t^2}=(\nabla\ln\mu)\cdot\nabla\vec{E}-(\nabla\vec{E})\cdot(\nabla\ln\mu)-\nabla[(\nabla\ln\epsilon)\cdot\vec{E}] \qquad v^2=\frac{1}{\mu\epsilon}$$



$$\vec{B}=\mu\vec{H} \qquad \vec{D}=\epsilon\vec{E}$$

理想绝缘介质中$\gamma=0$, 考虑自由电荷和传导电流为零的情况,

$$\nabla\times\vec{E}=-\frac{\partial\vec{B}}{\partial t} \quad \nabla\times\vec{B}-(\nabla\ln\mu)\times\vec{B}=\mu\epsilon\frac{\partial\vec{E}}{\partial t} \quad \nabla\cdot\vec{E}=-(\nabla\ln\epsilon)\cdot\vec{E} \quad \nabla\cdot\vec{B}=0$$

$$\nabla^2\vec{E}-\frac{1}{v^2}\frac{\partial^2\vec{E}}{\partial t^2}=(\nabla\ln\mu)\cdot\nabla\vec{E}-(\nabla\vec{E})\cdot(\nabla\ln\mu)-\nabla[(\nabla\ln\epsilon)\cdot\vec{E}] \qquad v^2=\frac{1}{\mu\epsilon}$$

$$\nabla^2\vec{B}=-\nabla(\nabla\cdot\vec{B})+\nabla^2\vec{B}=-\nabla\times(\nabla\times\vec{B})=-\nabla\times[(\mu\epsilon\frac{\partial\vec{E}}{\partial t})+\nabla(\ln\mu)\times\vec{B}]$$

$$=\mu\epsilon\frac{\partial^2}{\partial t^2}\vec{B}-\nabla(\mu\epsilon)\times\frac{\partial\vec{E}}{\partial t}-\nabla\times[\nabla(\ln\mu)\times\vec{B}]$$

$$=\mu\epsilon\frac{\partial^2}{\partial t^2}\vec{B}-\nabla\ln(\mu\epsilon)\times[\nabla\times\vec{B}-(\nabla\ln\mu)\times\vec{B}]-\nabla\times[\nabla(\ln\mu)\times\vec{B}]$$

$$\nabla^2\vec{B}-\frac{1}{v^2}\frac{\partial^2\vec{B}}{\partial t^2}=-\nabla\ln(\mu\epsilon)\times[\nabla\times\vec{B}-(\nabla\ln\mu)\times\vec{B}]-\nabla\times[\nabla(\ln\mu)\times\vec{B}]$$

$c^2=\frac{1}{\mu_0\epsilon_0}$把光与静磁和静电联系起来了，光的本质是电磁波！

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理想绝缘介质中的波动方程及平面电磁波解：真空中波动方程

$$(\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2})\vec{B} = 0 \qquad (\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2})\vec{E} = 0$$

从量子的角度看：

- $-i\hbar\nabla \Leftrightarrow \vec{p}$ $\qquad i\hbar\frac{\partial}{\partial t} \Leftrightarrow E$
- $\nabla^2 \Leftrightarrow -\frac{1}{\hbar^2}\vec{p}\cdot\vec{p}$ $\qquad -\frac{1}{c^2}\frac{\partial^2}{\partial t^2} \Leftrightarrow \frac{1}{c^2\hbar^2}E^2$
- $(-\vec{p}\cdot\vec{p} + E^2/c^2)\vec{B} = 0$ $\qquad (-\vec{p}\cdot\vec{p} + E^2/c^2)\vec{E} = 0$
- $E^2/c^2 - \vec{p}\cdot\vec{p} = m^2c^2$ $\qquad \Rightarrow \qquad m^2_{光子} = 0$

在非相对论的牛顿力学中是不允许存在零质量粒子的！

如果光子有非零静止质量$m^2_{光子}$：

$$(-\vec{p}\cdot\vec{p} + E^2/c^2 - m^2_{光子}c^2)\vec{B} = 0 \qquad (-\vec{p}\cdot\vec{p} + E^2/c^2 - m^2_{光子}c^2)\vec{E} = 0$$

$$(\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \frac{m^2_{光子}c^2}{\hbar^2})\vec{B} = 0 \qquad (\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \frac{m^2_{光子}c^2}{\hbar^2})\vec{E} = 0$$



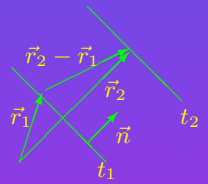
均匀介质：$\vec{E}(\vec{r},t)=\vec{E}(t-\frac{\vec{n}\cdot\vec{r}}{v})$　$\vec{B}(\vec{r},t)=\vec{B}(t-\frac{\vec{n}\cdot\vec{r}}{v})$　$\vec{n}\cdot\vec{n}=1$

$$\nabla\vec{E}(t-\frac{\vec{n}\cdot\vec{r}}{v})=\nabla(t-\frac{\vec{n}\cdot\vec{r}}{v})\frac{\partial\vec{E}(t')}{\partial t'}\Big|_{t'=t-\frac{\vec{n}\cdot\vec{r}}{v}}=-\frac{\vec{n}}{v}\frac{\partial\vec{E}}{\partial t}\underbrace{\frac{\partial t}{\partial t'}\Big|_{t'=t-\frac{\vec{n}\cdot\vec{r}}{v}}}_{=1}$$

$$\nabla\cdot\nabla\vec{E}=-\frac{\vec{n}}{v}\cdot\nabla\frac{\partial\vec{E}}{\partial t}=-\frac{\vec{n}}{v}\cdot\nabla(t-\frac{\vec{n}\cdot\vec{r}}{v})\frac{\partial^2\vec{E}}{\partial t^2}=\frac{1}{v^2}\frac{\partial^2\vec{E}}{\partial t^2}$$

- 对给定时刻,等值面为与$\vec{n}$垂直的平面：$\vec{r}_\perp\cdot\vec{n}=0$　$\vec{n}\cdot\vec{r}=\vec{n}\cdot(\vec{r}+\vec{r}_\perp)$

$$t-\frac{\vec{n}\cdot\vec{r}}{v}=t-\frac{\vec{n}\cdot(\vec{r}+\vec{r}_\perp)}{v}$$



- 等值面以速度$\vec{v}$沿$\vec{n}$方向运动：两个时刻$t_1<t_2$，等值面由$\vec{r}_1$移到$\vec{r}_2$,

$$t_1-\frac{\vec{n}\cdot\vec{r}_1}{v}=t_2-\frac{\vec{n}\cdot\vec{r}_2}{v}\quad\rightarrow\quad v=\vec{n}\cdot(\frac{\vec{r}_2-\vec{r}_1}{t_2-t_1})$$





$$\nabla \times \vec{\mathbf{E}} = \nabla(\mathbf{t} - \frac{\vec{\mathbf{r}} \cdot \vec{\mathbf{n}}}{\mathbf{v}}) \times \frac{\partial \vec{\mathbf{E}}}{\partial \mathbf{t}} = -\frac{\vec{\mathbf{n}}}{\mathbf{v}} \times \frac{\partial \vec{\mathbf{E}}}{\partial \mathbf{t}} = -\frac{\partial \vec{\mathbf{B}}}{\partial \mathbf{t}}$$

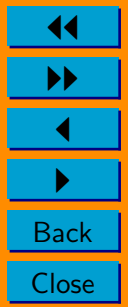
$$\nabla \times \vec{\mathbf{B}} = \nabla(\mathbf{t} - \frac{\vec{\mathbf{r}} \cdot \vec{\mathbf{n}}}{\mathbf{v}}) \times \frac{\partial \vec{\mathbf{B}}}{\partial \mathbf{t}} = -\frac{\vec{\mathbf{n}}}{\mathbf{v}} \times \frac{\partial \vec{\mathbf{B}}}{\partial \mathbf{t}} = \mu \nabla \times \vec{\mathbf{H}} = \mu \frac{\partial \vec{\mathbf{D}}}{\partial \mathbf{t}}$$

$$= \mu\epsilon \frac{\partial \vec{\mathbf{E}}}{\partial \mathbf{t}} = \frac{1}{\mathbf{v^2}} \frac{\partial \vec{\mathbf{E}}}{\partial \mathbf{t}}$$

$$\frac{\partial}{\partial \mathbf{t}}(\mathbf{v}\vec{\mathbf{B}} - \vec{\mathbf{n}} \times \vec{\mathbf{E}}) = \mathbf{0} \rightarrow \quad \mathbf{v}\vec{\mathbf{B}} = \vec{\mathbf{n}} \times \vec{\mathbf{E}} + \text{时间无关项}$$

$$\frac{\partial}{\partial \mathbf{t}}(\vec{\mathbf{n}} \times \mathbf{v}\vec{\mathbf{B}} + \vec{\mathbf{E}}) = \mathbf{0} \rightarrow \quad \vec{\mathbf{E}} = -\vec{\mathbf{n}} \times \mathbf{v}\vec{\mathbf{B}} + \text{时间无关项}$$

$$= -\vec{\mathbf{v}} \times \vec{\mathbf{B}} + \text{时间无关项}$$



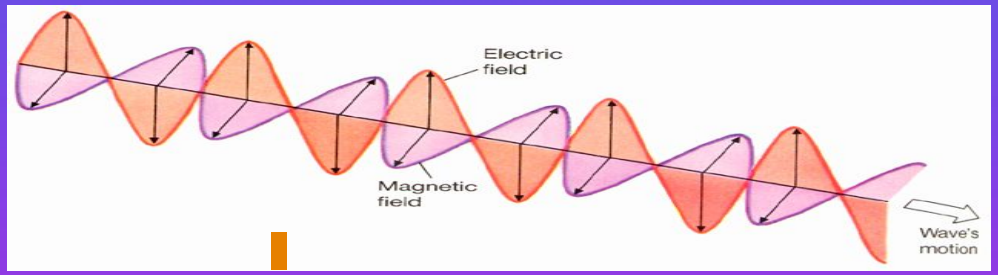
理想绝缘介质中的波动方程及平面电磁波解： 平面波解

$v\vec{B} = \vec{n} \times \vec{E}$ + 时间无关项　　$\vec{E} = -\vec{v} \times \vec{B}$ + 时间无关项

$$0 = \nabla \cdot \vec{E} = \nabla(t - \frac{\vec{r} \cdot \vec{n}}{v}) \cdot \frac{\partial \vec{E}}{\partial t} = -\frac{\partial}{\partial t}(\frac{\vec{n}}{v} \cdot \vec{E}) \rightarrow \vec{n} \cdot \vec{E} = \text{时间无关项}$$

$$0 = \nabla \cdot \vec{B} = \nabla(t - \frac{\vec{r} \cdot \vec{n}}{v}) \cdot \frac{\partial \vec{B}}{\partial t} = -\frac{\partial}{\partial t}(\frac{\vec{n}}{v} \cdot \vec{B}) \rightarrow \vec{n} \cdot \vec{B} = \text{时间无关项}$$

略去时间无关项(静态电磁场)：$\vec{E}, \vec{B}, \vec{n}$ 服从右手关系！　$vB = E$



$$W_e = \frac{1}{2}\epsilon E^2 = \frac{1}{2}\epsilon v^2 B^2 = \frac{1}{2\mu}B^2 = W_m$$

$$\vec{S} = \vec{E} \times \vec{H} = \vec{E} \times \frac{\vec{B}}{\mu} = -(\vec{v} \times \vec{B}) \times \frac{\vec{B}}{\mu} = \vec{v}\frac{B^2}{\mu} = \vec{v}(W_e + W_m)$$

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导体与绝缘介质的差别在于对导体存在欧姆定律$\vec{j}=\gamma\vec{E}$. $\gamma=0$则回到理想绝缘介质.

$$\frac{\partial\rho}{\partial t}=-\nabla\cdot\vec{j}=-\nabla\cdot(\gamma\vec{E})=-\frac{\gamma}{\epsilon_0}\rho$$

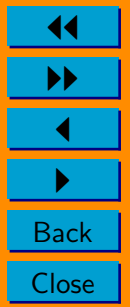
$$\rho(t)=\rho(0)e^{-\frac{\gamma}{\epsilon_0}t}$$

在导体中,电荷密度是很快衰减的!
(在静电时,导体内部不带电,电荷只分布在表面上).

讨论如下两种情况:

$$\begin{cases}\gamma\sim 0 & \rho(0)=0\\ \gamma\gg\epsilon\omega & \omega:\text{电磁波的角频率}\end{cases}$$

近似地取 $\rho(t)=0$



定态波动方程及平面波解：定态电磁波

$\vec{D}(\omega)=\epsilon(\omega)\vec{E}(\omega)\qquad \vec{B}(\omega)=\mu(\omega)\vec{H}(\omega)\qquad \vec{j}(\omega)=\gamma(\omega)\vec{E}(\omega)$

定态电磁波：$\vec{E}(\vec{r},t)=\vec{E}(\vec{r})e^{-i\omega t}\qquad \vec{B}(\vec{r},t)=\vec{B}(\vec{r})e^{-i\omega t}$ 非定态一般不满足波动方程

$$\begin{cases}\nabla\times\vec{E}=i\omega\vec{B}\\ \nabla\times\vec{H}=\vec{j_c}-i\omega\vec{D}\\ \nabla\cdot\vec{D}=0\\ \nabla\cdot\vec{B}=0\end{cases}$$ 代入物质电磁方程 $$\begin{cases}\nabla\times\vec{E}=i\omega\vec{B}\\ \nabla\times\vec{B}=\mu(\gamma-i\omega\epsilon)\vec{E}\\ \nabla\cdot\vec{E}=0\\ \nabla\cdot\vec{B}=0\end{cases}$$

$$\nabla^2\vec{E}=\nabla^2\vec{E}-\nabla(\nabla\cdot\vec{E})=-\nabla\times(\nabla\times\vec{E})=-i\omega\nabla\times\vec{B}$$
$$=-i\omega\mu(\gamma-i\omega\epsilon)\vec{E}$$
$$\nabla^2\vec{B}=\nabla^2\vec{B}-\nabla(\nabla\cdot\vec{B})=-\nabla\times(\nabla\times\vec{B})=-\mu(\gamma-i\omega\epsilon)\nabla\times\vec{E}$$
$$=-i\omega\mu(\gamma-i\omega\epsilon)\vec{B}$$

$$\begin{cases}(\nabla^2+k^2)\vec{E}=0\\ \nabla\cdot\vec{E}=0\\ \vec{B}=-\frac{i}{\omega}\nabla\times\vec{E}\end{cases}$$ $\vec{E},\vec{B}$相互不独立 $\leftarrow\!-\!-\!-\!-\!-\!-\!-\!\rightarrow$ $k^2\equiv\omega^2\mu\epsilon+i\omega\mu\gamma$ $$\begin{cases}(\nabla^2+k^2)\vec{B}=0\\ \nabla\cdot\vec{B}=0\\ \vec{E}=\frac{1}{\mu(\gamma-i\omega\epsilon)}\nabla\times\vec{B}\end{cases}$$

导体的γ可以吸收到ϵ的虚部中去！ $\epsilon\Rightarrow\epsilon+i\frac{\gamma}{\omega}$。电导率可从介电常数的虚部抽取！

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$$\vec{E}(\vec{r})=\vec{E}_0e^{i\vec{k}\cdot\vec{r}} \qquad \vec{E}(\vec{r},t)=\vec{E}_0e^{i(\vec{k}\cdot\vec{r}-\omega t)} \qquad \vec{k}\cdot\vec{k}\equiv k^2=\omega^2\mu\epsilon+i\omega\mu\gamma$$

$$\vec{B}(\vec{r})=\vec{B}_0e^{i\vec{k}\cdot\vec{r}} \qquad \vec{B}(\vec{r},t)=\vec{B}_0e^{i(\vec{k}\cdot\vec{r}-\omega t)}$$

量子对应：$\vec{k}\sim\vec{p}/\hbar \quad \omega\sim E/\hbar \quad \Rightarrow \quad -i\hbar\nabla\sim\vec{p} \quad i\hbar\frac{\partial}{\partial t}\sim E$ 符号选择？

$\vec{k}$： 波矢 $\vec{k}=\vec{k}_R+i\vec{k}_I$. $\quad t-\frac{\vec{k}}{\omega}\cdot\vec{r}\overset{\frac{k}{\omega}\big|_{\gamma=0}=\sqrt{\mu\epsilon}=\frac{1}{v},\ \vec{n}\equiv\frac{\vec{k}}{k}}{=\!=\!=\!=\!=\!=\!=\!=\!=\!=\!=\!=\!=\!=\!=\!=}t-\frac{\vec{n}\cdot\vec{r}}{v}$

$\vec{E}_0,\vec{B}_0$： 复振幅 $E_{0n}=\|E_{0n}\|e^{i\delta_{En}} \quad B_{0n}=\|B_{0n}\|e^{i\delta_{Bn}}$.

$$\vec{E}(\vec{r},t)=\sum_{n=1}^{3}\vec{e}_n\|E_{0n}\|e^{-\vec{k}_I\cdot\vec{r}}e^{i(\vec{k}_R\cdot\vec{r}-\omega t+\delta_{En})}$$

$$\vec{B}(\vec{r},t)=\sum_{n=1}^{3}\vec{e}_n\|B_{0n}\|e^{-\vec{k}_I\cdot\vec{r}}e^{i(\vec{k}_R\cdot\vec{r}-\omega t+\delta_{Bn})}$$

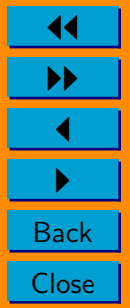
$\|E_{0n}\|e^{-\vec{k}_I\cdot\vec{r}}$ 是电场强度在n方向分量的振幅.

$\|B_{0n}\|e^{-\vec{k}_I\cdot\vec{r}}$ 是磁感应强度在n方向分量的振幅.

$\vec{k}_R\cdot\vec{r}-\omega t+\delta_{En}$ 是电场强度在n方向分量的位相.

$\vec{k}_R\cdot\vec{r}-\omega t+\delta_{Bn}$ 是磁感应强度在n方向分量的位相.





$$\vec{E}(\vec{r},t)=\sum_{n=1}^{3}\vec{e}_n\|E_{0n}\|e^{-\vec{k}_I\cdot\vec{r}}e^{i(\vec{k}_R\cdot\vec{r}-\omega t+\delta_{En})}$$

- 在给定时刻t,等相面与$\vec{k}_R$垂直.

  $\vec{k}_R\cdot\vec{r}=\vec{k}_R\cdot(\vec{r}+\vec{r}_\perp)$　　$\vec{r}_\perp\cdot\vec{k}_R=0$

- 等相面以速率$v_\phi=\omega/k_R$沿$\vec{k}_R$方向运动，$v_\phi=\frac{(\vec{r}_2-\vec{r}_1)}{t_2-t_1}\cdot\frac{\vec{k}_R}{k_R}$.

$$-\omega t_1+\vec{k}_R\cdot\vec{r}_1=-\omega t_2+\vec{k}_R\cdot\vec{r}_2$$

$$\rightarrow\ 1=\frac{\vec{k}_R}{\omega}\cdot\frac{(\vec{r}_2-\vec{r}_1)}{t_2-t_1}=\frac{k_R}{\omega}\frac{\vec{k}_R}{k_R}\cdot\frac{(\vec{r}_2-\vec{r}_1)}{t_2-t_1}=\frac{k_R}{\omega}v_\phi$$

- 当$\vec{k}_I=0$时,等相面就是等值面.
- 当$\vec{k}_I\neq 0$时,等值区域由 $\vec{k}_R\cdot\vec{r}=$常数和 $\vec{k}_I\cdot\vec{r}=$常数联合求解决定,一般是一条线.沿$\vec{k}_I$方向，振幅衰减最快，反$\vec{k}_I$方向，振幅增强最快 .



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- 波矢：
  - $\vec{k}_R$的方向反映的是位相的传播方向. $\vec{k}_I$的方向反映的是振幅的最大衰减方向.

$$k^2=\omega^2\mu\epsilon+i\omega\mu\gamma=\vec{k}\cdot\vec{k}=k_R^2-k_I^2+2i\vec{k}_R\cdot\vec{k}_I$$ 复矢量的模是$\vec{k}\cdot\vec{k}$! 不是$\vec{k}\cdot\vec{k}^*$哟！

$$k_R^2-k_I^2=\mu\epsilon\omega^2 \qquad \vec{k}_R\cdot\vec{k}_I=\frac{1}{2}\omega\mu\gamma$$ k, k_R, k_I 的符号如何取？

$\vec{k}_R\cdot\vec{k}_I>0 \Rightarrow$ $\omega\mu\gamma>0$时沿位相传播方向波总是衰减的！

$\vec{k}_R\cdot\vec{k}_I \overset{\text{绝缘介质}}{=\!=\!=\!=} 0 \Rightarrow$ 绝缘介质有可能有衰减！在自由空间通过傅立叶展开仍用无衰减的波描写

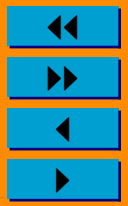
绝缘介质若有衰减，最大衰减方向与传播方向相互垂直！

但在边界上边界条件限制不允许再用傅立叶展开因而出现明显衰减

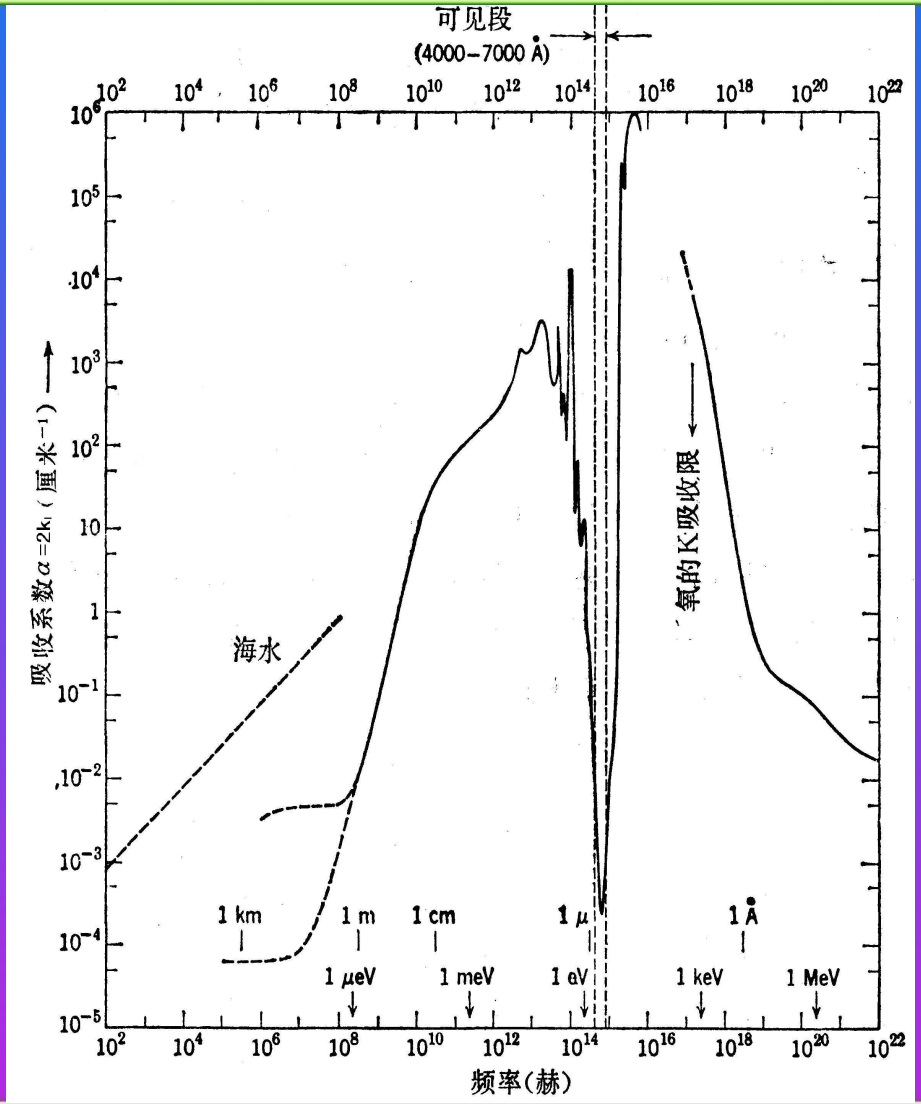
  - k_I代表电磁波振幅的衰减.也反映了物质对电磁波的吸收. 对绝缘介质,$\gamma=0, k_I=0$,对绝缘介质,电磁波无吸收,电磁能量不耗散.
  - 波幅降至原值$1/e$的传播距离为穿透深度δ： $1=\vec{k}_I\cdot\delta\vec{k}_R/k_R$

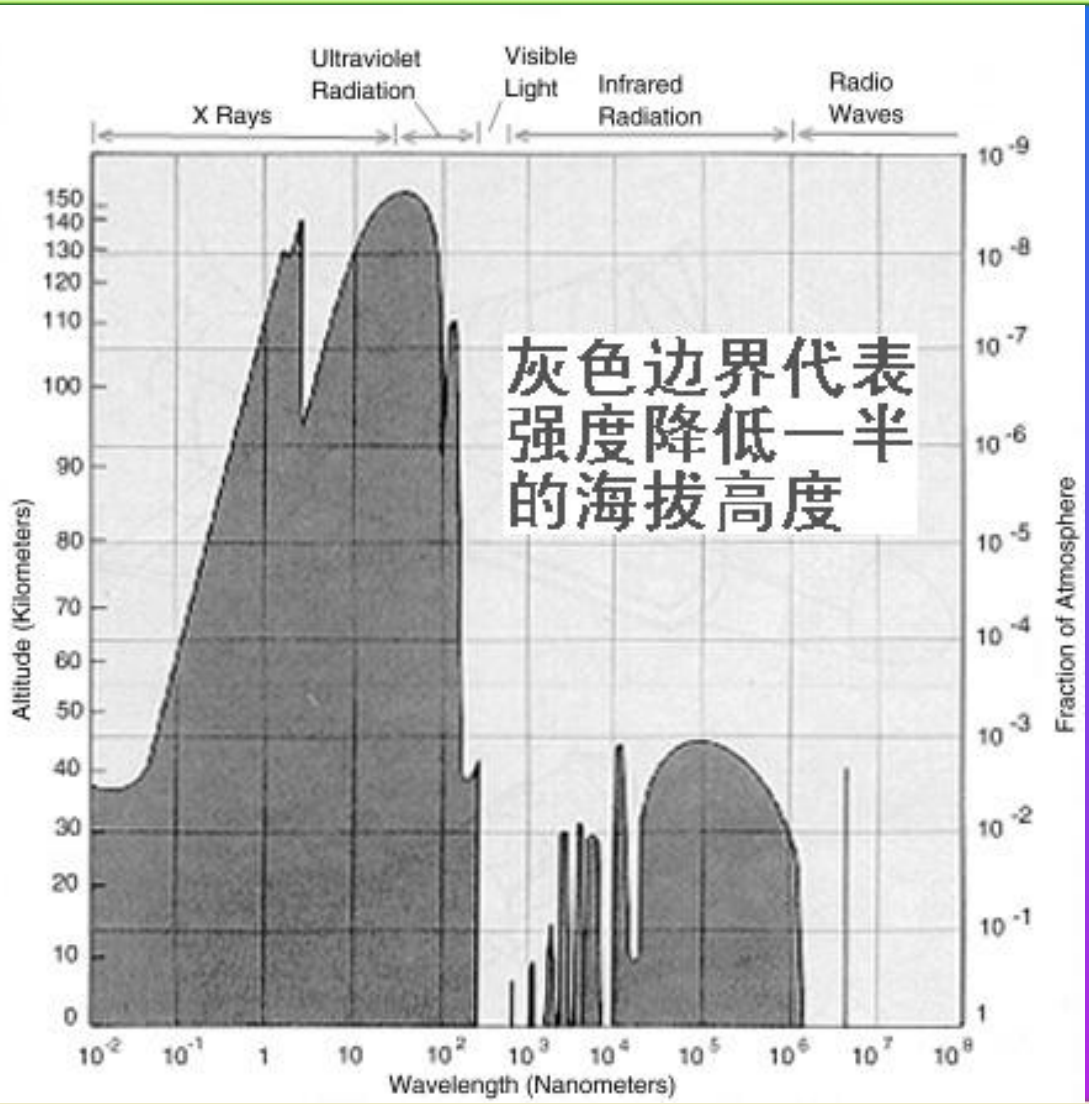
$$\delta=\frac{k_R}{\vec{k}_I\cdot\vec{k}_R}=\frac{2k_R}{\omega\mu\gamma}$$

  - k_I与ω有关,反映这种物质 对某些频率的电磁波吸收的强弱.



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可见段
(4000–7000 Å)
吸收系数 $\alpha = 2k_i$（厘米$^{-1}$）
海水
氧的K吸收限
1 km
1 m
1 cm
1 μ
1 Å
1 μeV
1 meV
1 eV
1 keV
1 MeV
频率(赫)


X Rays
Ultraviolet Radiation
Visible Light
Infrared Radiation
Radio Waves
灰色边界代表强度降低一半的海拔高度
Altitude (Kilometers)
Fraction of Atmosphere
Wavelength (Nanometers)



- 振幅：复振幅$\vec{E}_0, \vec{B}_0$满足 $\vec{k}\cdot\vec{E}_0 = 0 \quad \vec{B}_0 = \frac{\vec{k}}{\omega}\times\vec{E}_0$
  - $k_I = 0$, $\vec{k}, \vec{E}_0, \vec{B}_0$ 成右手关系, $B_0 = \frac{k}{\omega}E_0$, $\vec{B}$与$\vec{E}$同位相.
  - $k_I \neq 0$, 右手关系不成立, $\vec{B}$与$\vec{E}$不同位相.

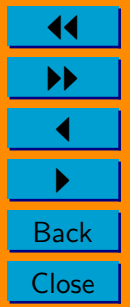
$$\vec{E}_0 = \vec{E}_{0R} + i\vec{E}_{0I} \qquad \vec{B}_0 = \vec{B}_{0R} + i\vec{B}_{0I}$$

$$\vec{k}_R\cdot\vec{E}_{0R} - \vec{k}_I\cdot\vec{E}_{0I} = 0 \qquad \vec{k}_R\cdot\vec{E}_{0I} + \vec{k}_I\cdot\vec{E}_{0R} = 0$$

$$\vec{B}_{0R} = \frac{\vec{k}_R}{\omega}\times\vec{E}_{0R} - \frac{\vec{k}_I}{\omega}\times\vec{E}_{0I} \qquad \vec{B}_{0I} = \frac{\vec{k}_R}{\omega}\times\vec{E}_{0I} + \frac{\vec{k}_I}{\omega}\times\vec{E}_{0R}$$

$$\vec{E}_{0R} = \sum_{n=1}^{3}\|E_{0n}\|\vec{e}_n\cos\delta_{En} \qquad \vec{E}_{0I} = \sum_{n=1}^{3}\|E_{0n}\|\vec{e}_n\sin\delta_{En}$$

$$\vec{B}_{0R} = \sum_{n=1}^{3}\|B_{0n}\|\vec{e}_n\cos\delta_{Bn} \qquad \vec{B}_{0I} = \sum_{n=1}^{3}\|B_{0n}\|\vec{e}_n\sin\delta_{Bn}$$





- 偏振：$\vec{k}$为实，z轴为$\vec{k}$方向(电磁波沿z轴正方向传播)

$$E_{0z}=0 \qquad \|E_{0x}\|\equiv A_x \qquad \|E_{0y}\|\equiv A_y \qquad \delta\equiv\delta_{Ex}-\delta_{Ey}$$

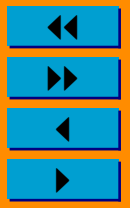
$$\vec{E}(\vec{r},t)=\|E_{0x}\|\cos(kz-\omega t+\delta_{Ex})\vec{e}_1+\|E_{0y}\|\cos(kz-\omega t+\delta_{Ey})\vec{e}_2$$

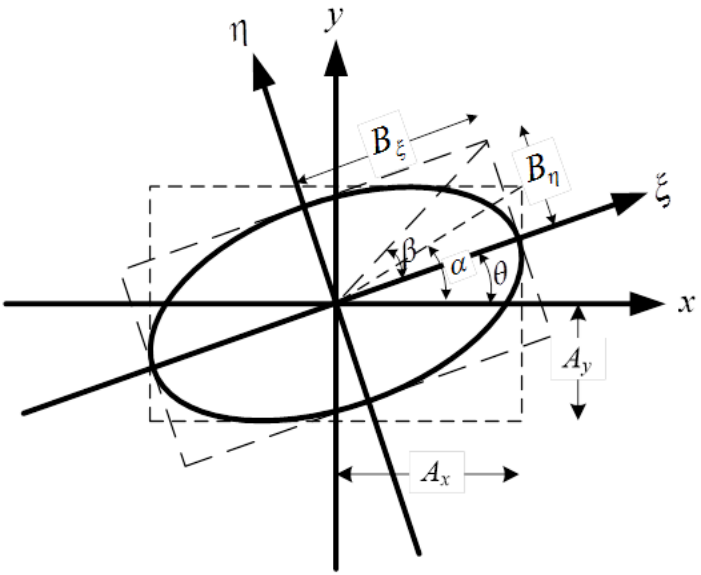
$$\left(\frac{E_x}{\|E_{0x}\|}\right)^2+\left(\frac{E_y}{\|E_{0y}\|}\right)^2-2\left(\frac{E_x}{\|E_{0x}\|}\right)\left(\frac{E_y}{\|E_{0y}\|}\right)\cos(\delta)=\sin^2\delta \qquad -\pi<\delta<\pi$$

$$\cos^2 X+\cos^2 Y-2\cos X\cos Y\cos(X-Y)=\sin^2(X-Y)$$

在某一z取定的xy平面上,电场强度矢量随时间不断变化

$$\begin{cases}\delta_{Ex}-\delta_{Ey}=0,\pi & \text{线偏振}\\ \|E_{0x}\|=\|E_{0y}\| & \begin{cases}\delta_{Ex}-\delta_{Ey}=\frac{\pi}{2} & \text{左旋}\\ \delta_{Ex}-\delta_{Ey}=\frac{3\pi}{2} & \text{右旋}\\ \delta_{Ex}-\delta_{Ey}\text{不固定} & \text{自然光}\end{cases}\\ \text{一般：椭圆}\end{cases}$$





$$\tan\alpha = A_y / A_x \qquad 0^\circ \le \alpha \le 90^\circ$$

$$\tan\beta = \pm B_\eta / B_\xi \qquad -45^\circ \le \beta \le 45^\circ$$

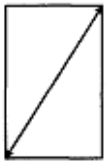
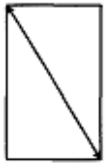
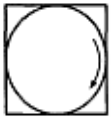
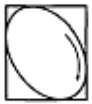
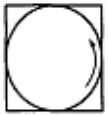
$$\begin{cases} \tan 2\theta = \tan 2\alpha \cos\delta \\ \sin 2\beta = \sin 2\alpha \sin\delta \\ B_\xi^2 + B_\eta^2 = A_x^2 + A_y^2 \end{cases}$$

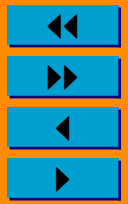









Ⅰ Ⅲ象限线	右旋椭圆			Ⅱ Ⅳ象限线	左旋椭圆		
							
$\tan\alpha = A_y / A_x \neq 1$							
$\delta = 0$	$0 < \delta < \pi/2$	$\delta = \pi/2$	$\pi/2 < \delta < \pi$	$\delta = \pi$	$-\pi < \delta < -\pi/2$	$\delta = -\pi/2$	$-\pi/2 < \delta < 0$
$\beta = 0$	$0° < \beta < 45°$			$\beta = 0$	$-45° < \beta < 0°$		
45°线偏振	右旋椭圆	右旋圆偏振	右旋椭圆	−45°线偏振	左旋椭圆	左旋圆偏振	左旋椭圆
							
$\tan\alpha = A_y / A_x = 1$							
$\delta = 0$	$0 < \delta < \pi/2$	$\delta = \pi/2$	$\pi/2 < \delta < \pi$	$\delta = \pi$	$-\pi < \delta < -\pi/2$	$\delta = -\pi/2$	$-\pi/2 < \delta < 0$
$\beta = 0$	$0° < \beta < 45°$	$\beta = 45°$	$0° < \beta < 45°$	$\beta = 0$	$-45° < \beta < 0°$	$\beta = -45°$	$-45° < \beta < 0°$



Back
Close



- 偏振: Stokes参数

$$\mathbf{E_{0z}=0} \qquad \vec{\mathbf{E}}(\vec{\mathbf{r}},\mathbf{t})=||\mathbf{E_{0x}}||\mathbf{e}^{\mathbf{i(kz-\omega t+\delta_{Ex})}}\vec{\mathbf{e}}_1+||\mathbf{E_{0y}}||\mathbf{e}^{\mathbf{i(kz-\omega t+\delta_{Ey})}}\vec{\mathbf{e}}_2$$

$$\mathbf{I=S_0}\equiv||\mathbf{E_x}||^2+||\mathbf{E_y}||^2=||\mathbf{E_{0x}}||^2+||\mathbf{E_{0y}}||^2$$

$$\mathbf{Q=S_1}\equiv||\mathbf{E_x}||^2-||\mathbf{E_y}||^2=||\mathbf{E_{0x}}||^2-||\mathbf{E_{0y}}||^2$$

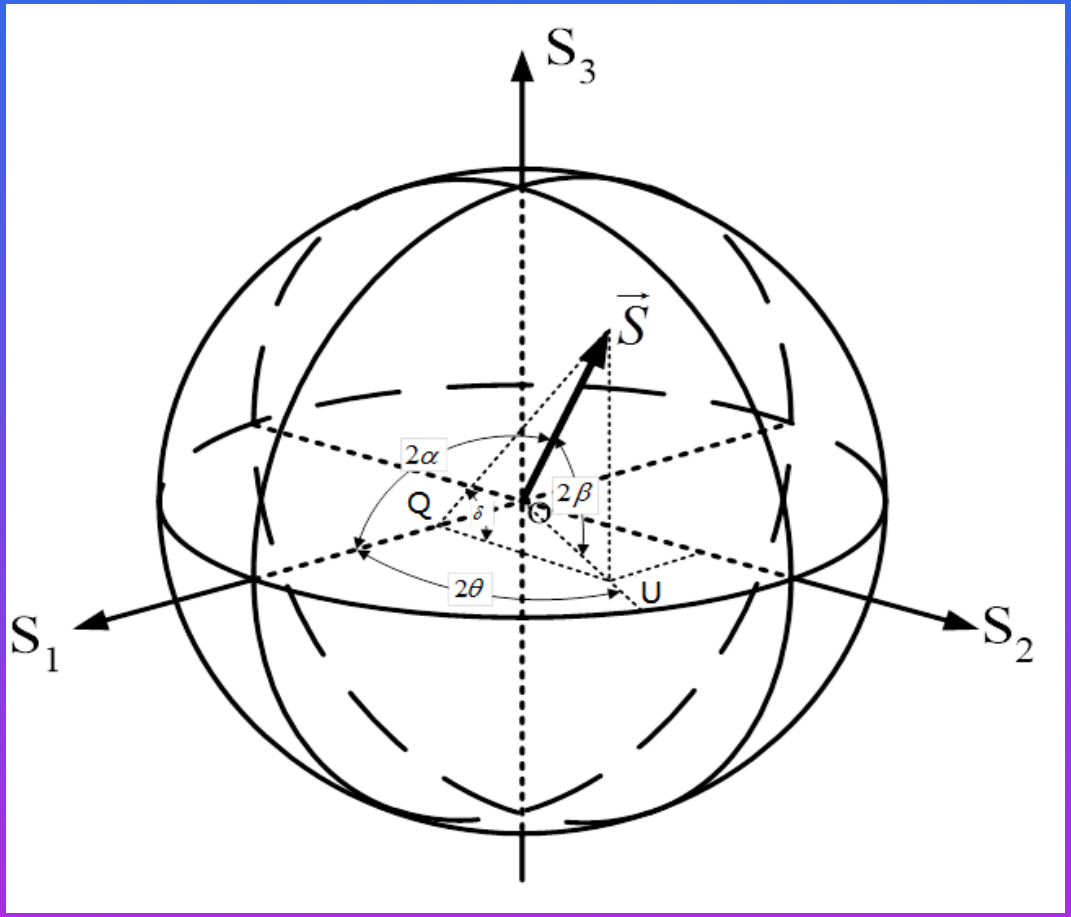
$$\mathbf{U=S_2}\equiv\mathbf{2}\mathrm{Re}(\mathbf{E_xE_y^*})=\mathbf{2}||\mathbf{E_{0x}E_{0y}}||\cos(\delta_{\mathbf{Ex}}-\delta_{\mathbf{Ey}})$$

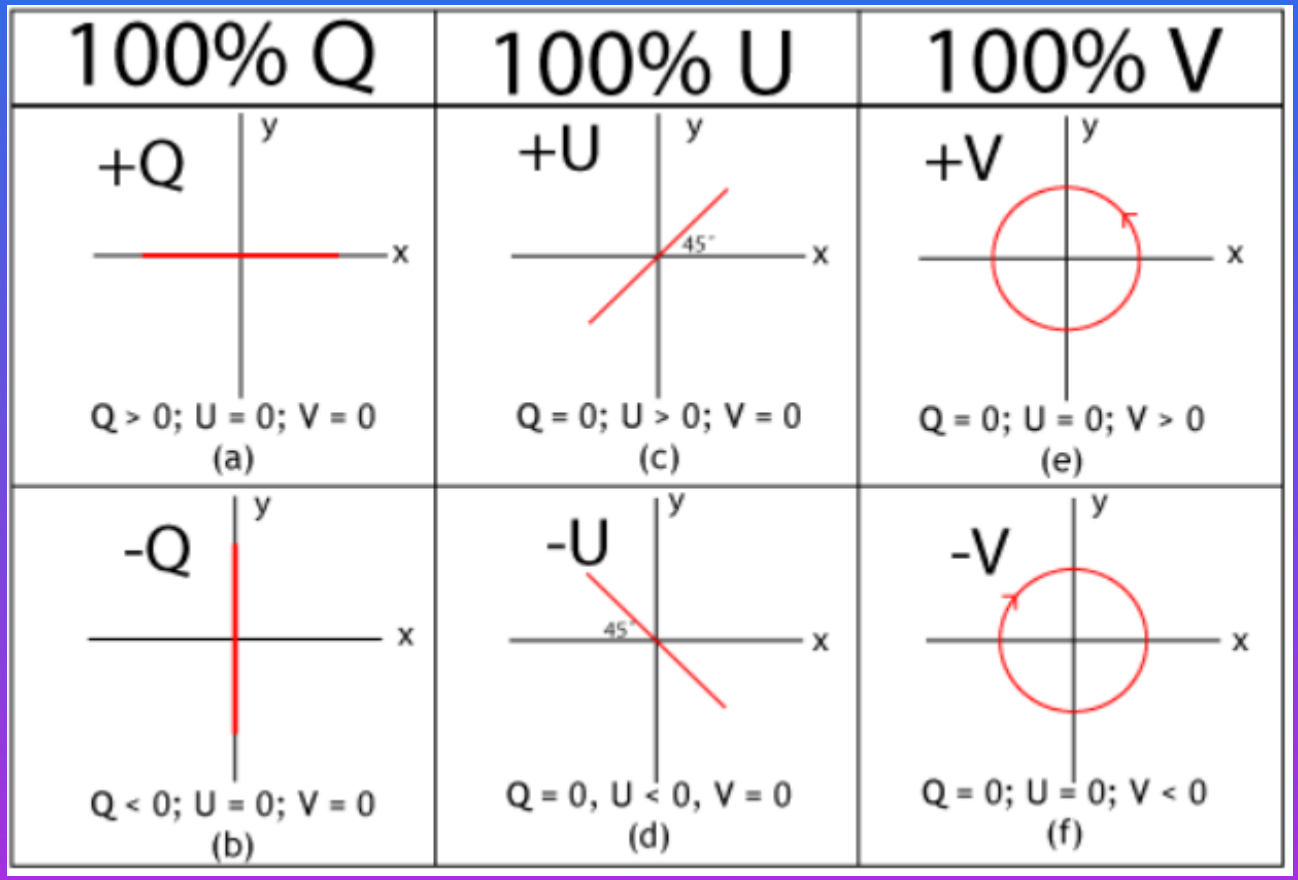
$$\mathbf{V=S_3}\equiv-\mathbf{2}\mathrm{Im}(\mathbf{E_xE_y^*})=-\mathbf{2}||\mathbf{E_{0x}E_{0y}}||\sin(\delta_{\mathbf{Ex}}-\delta_{\mathbf{Ey}})$$

$$\mathbf{Q^2+U^2+V^2=I^2} \qquad \mathbf{S_1^2+S_2^2+S_3^2=S_0^2}$$

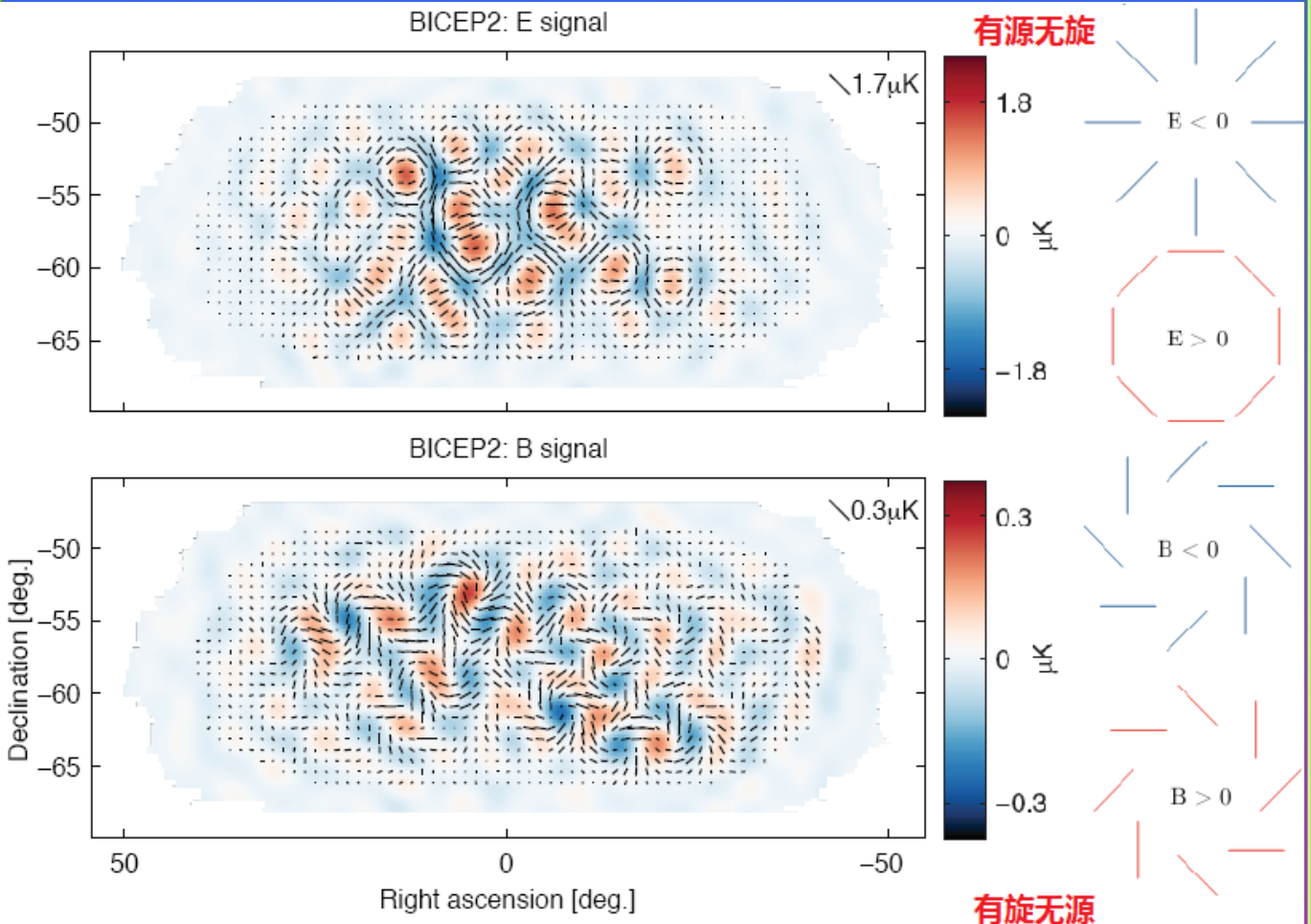
Poincaré球!

Back
Close


S3
S1
S2
$\vec{S}$
2α
2β
2θ
δ
Q
O
U



$$\mathbf{Q}=||\mathbf{E_{0x}}||^2-||\mathbf{E_{0y}}||^2 \quad \mathbf{U}=2||\mathbf{E_{0x}E_{0y}}||\cos(\delta_{\mathbf{Ex}}-\delta_{\mathbf{Ey}}) \quad \mathbf{V}=-2||\mathbf{E_{0x}E_{0y}}||\sin(\delta_{\mathbf{Ex}}-\delta_{\mathbf{Ey}})$$


BICEP2: E signal
1.7μK
−50
−55
−60
−65
1.8
0
−1.8
μK
有源无旋
E < 0
E > 0
BICEP2: B signal
0.3μK
0.3
0
−0.3
μK
B < 0
B > 0
Declination [deg.]
50
0
−50
Right ascension [deg.]
有旋无源



右照片利用偏振片，被窗户玻璃反射的强光被过滤掉可清楚看到玻璃后的窗帘。



散射可起偏。蓝天是太阳光散射结果，不同角度天空存在不同程度偏振，与太阳成90度方向是100%线偏振。对摄影有用，对比上面两张图。



- 能量密度，能流密度：

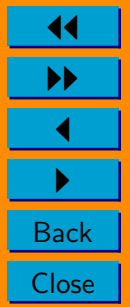
$$W_e = \frac{1}{2}\epsilon \vec{E}_R \cdot \vec{E}_R \qquad W_m = \frac{1}{2\mu}\vec{B}_R \cdot \vec{B}_R \qquad \vec{S} = \frac{1}{\mu}\vec{E}_R \times \vec{B}_R$$

$$\vec{E}_R \cdot \vec{E}_R = \sum_{n=1}^{3} E_{Rn}E_{Rn} = \sum_{n=1}^{3} \|E_{0n}\|^2 e^{-2\vec{k}_I \cdot \vec{r}} \cos^2(\vec{k}_R \cdot \vec{r} - \omega t + \delta_{En})$$

$$\vec{B}_R \cdot \vec{B}_R = \sum_{n=1}^{3} B_{Rn}B_{Rn} = \sum_{n=1}^{3} \|B_{0n}\|^2 e^{-2\vec{k}_I \cdot \vec{r}} \cos^2(\vec{k}_R \cdot \vec{r} - \omega t + \delta_{Bn})$$

$$\vec{E}_R \times \vec{B}_R = \sum_{n,m,p=1}^{3} \epsilon_{nmp} E_{Rn} B_{Rm} \vec{e}_p$$

$$= \sum_{n,m,p=1}^{3} \epsilon_{nmp} \|E_{0n}\| \|B_{0m}\| e^{-2\vec{k}_I \cdot \vec{r}} \vec{e}_p \cos(\vec{k}_R \cdot \vec{r} - \omega t + \delta_{En}) \times \cos(\vec{k}_R \cdot \vec{r} - \omega t + \delta_{Bn})$$



- 能量密度，能流密度：

$$\overline{\cos^2(\vec{k}_R\cdot\vec{r}-\omega t+\delta_{En})}=\frac{1}{2}\overline{[1+\cos 2(\vec{k}_R\cdot\vec{r}-\omega t+\delta_{En})]}=\frac{1}{2}$$

$$\overline{\cos^2(\vec{k}_R\cdot\vec{r}-\omega t+\delta_{Bn})}=\frac{1}{2}\overline{[1+\cos 2(\vec{k}_R\cdot\vec{r}-\omega t+\delta_{Bn})]}=\frac{1}{2}$$

$$\overline{\cos(\vec{k}_R\cdot\vec{r}-\omega t+\delta_{En})\cos(\vec{k}_R\cdot\vec{r}-\omega t+\delta_{Bn})}$$
$$=\frac{1}{2}\overline{[\cos(2\vec{k}_R\cdot\vec{r}-2\omega t+\delta_{En}+\delta_{Bn})+\cos(\delta_{En}-\delta_{Bn})]}=\frac{1}{2}\cos(\delta_{En}-\delta_{Bn})$$

$$\overline{\vec{E}_R\cdot\vec{E}_R}=\frac{1}{2}\sum_{n=1}^{3}\|E_{0n}^2\|e^{-2\vec{k}_I\cdot\vec{r}}=\frac{1}{2}Re(\vec{E}\cdot\vec{E}^*)$$

$$\overline{\vec{B}_R\cdot\vec{B}_R}=\frac{1}{2}\sum_{n=1}^{3}\|B_{0n}^2\|e^{-2\vec{k}_I\cdot\vec{r}}=\frac{1}{2}Re(\vec{B}\cdot\vec{B}^*)$$

$$\overline{\vec{E}_R\times\vec{B}_R}=\frac{1}{2}\sum_{n,m,p=1}^{3}\epsilon_{nmp}\|E_{0n}\|\|B_{0m}\|\vec{e}_p e^{-2\vec{k}_I\cdot\vec{r}}\cos(\delta_{En}-\delta_{Bn})$$
$$=\frac{1}{2}Re(\vec{E}\times\vec{B}^*)$$



- 能量密度，能流密度：

$$\overline{W_e} = \frac{1}{2}\epsilon\overline{\vec{E}_R \cdot \vec{E}_R} = \frac{1}{4}\epsilon Re(\vec{E}\cdot\vec{E}^*)$$

$$\overline{W_m} = \frac{1}{2\mu}\overline{\vec{B}_R \cdot \vec{B}_R} = \frac{1}{4\mu}Re(\vec{B}\cdot\vec{B}^*) = \frac{1}{4\mu}Re\left[(\frac{\vec{k}}{\omega}\times\vec{E})\cdot(\frac{\vec{k}^*}{\omega}\times\vec{E}^*)\right]$$

$$= \frac{1}{4\mu\omega^2}Re\left[(\vec{k}\cdot\vec{k}^*)(\vec{E}\cdot\vec{E}^*) - (\vec{k}\cdot\vec{E}^*)(\vec{k}^*\cdot\vec{E})\right]$$

对绝缘体：$\vec{k} = \vec{k}^*,\ \vec{k}\cdot\vec{E} = 0 \rightarrow\ W_e = W_m$　对导体：$W_e \neq W_m$

$$\vec{S} = \frac{\overline{\vec{E}_R \times \vec{B}_R}}{\mu} = \frac{1}{2\mu}Re(\vec{E}\times\vec{B}^*)$$

$$= \frac{1}{2\mu}Re(\vec{E}^*\times\vec{B}) = \frac{1}{2\mu}Re[\vec{E}^*\times(\frac{\vec{k}}{\omega}\times\vec{E})]$$

$$= \frac{1}{2\mu\omega}Re[\vec{k}(\vec{E}\cdot\vec{E}^*) - \vec{E}(\vec{k}\cdot\vec{E}^*)]$$



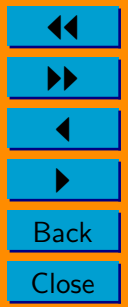







自然界 弥散能量 的构成:

- 在宏观或宇观世界中，能量可以是弥散的也可以由非零质量的实物粒子携带
- 一团弥散的纯能量定义为不包含实物粒子粒子物理标准模型中的物质粒子的能量构成形式
- 它最可能通过爆炸产生，特别可能由一对正反粒子湮灭产生
- 因为由零质量粒子组成，这团弥散能量一定是以光速运动着的
- 弥散纯能量的最大可能构成就是电磁波
- 还有可能包括引力波，但它太弱 其它媒介：胶子被囚禁在强子中；中性玻色子、希格斯太重
- 还有包括中微子，但现在中微子因有质量应被归为实物粒子
- 电磁波是自然界最为主要的弥散纯能量包括常说的辐射的构成方式 ！



入射波：$\vec{A}_1 = \vec{A}_{10} e^{i(\vec{k}_1 \cdot \vec{r} - \omega t)}$　　透射波：$\vec{A}_2 = \vec{A}_{20} e^{i(\vec{k}_2 \cdot \vec{r} - \omega_2 t)}$

反射波：$\vec{A}_3 = \vec{A}_{30} e^{i(\vec{k}_3 \cdot \vec{r} - \omega_3 t)}$　　$\vec{A} \subset (\vec{E}, \vec{H})$

$$\vec{k}_1 = k_1(\sin\theta_1 \vec{e}_1 + \cos\theta_1 \vec{e}_3) \quad \vec{k}_3 = k_{3x}\vec{e}_1 + k_{3y}\vec{e}_2 + k_{3z}\vec{e}_3 \quad \vec{k}_2 = \vec{k}_{2R} + i\vec{k}_{2I}$$

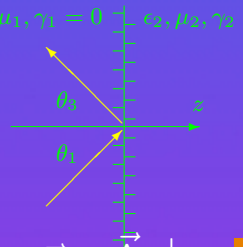
$$k_1 = \frac{\omega}{v} = \sqrt{\mu_1\epsilon_1}\omega \quad k_3 = \frac{\omega_3}{v} = \sqrt{\mu_1\epsilon_1}\omega_3 \quad \begin{cases} k_{2R}^2 - k_{2I}^2 = \mu_2\epsilon_2\omega_2^2 \\ \vec{k}_{2R} \cdot \vec{k}_{2I} = \frac{1}{2}\omega_2\mu_2\gamma_2 \end{cases}$$

已知：$\vec{k}_1$，ω，$\vec{A}_{10}$，μ_1，ϵ_1，γ_1，μ_2，ϵ_2，γ_2

需求：$\vec{k}_2$，$\vec{k}_3$，ω_2，ω_3，$\vec{A}_{20}$若有零解证明可不存在，$\vec{A}_{30}$

边条件：$\vec{n} \times (\vec{E}_2 - \vec{E}_1) = 0$，$\vec{n} \times (\vec{H}_2 - \vec{H}_1) = \vec{i}_c$

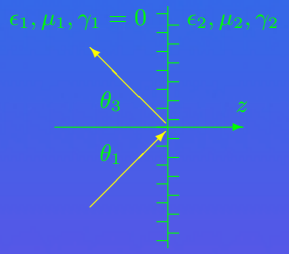
$\vec{n} \cdot (\vec{D}_2 - \vec{D}_1) = \sigma_f$，$\vec{n} \cdot (\vec{B}_2 - \vec{B}_1) = 0$



$\vec{i}_c = 0$←有限J←有限E，σ_f待定其变化直接导致法向电流，$\vec{n} \times (\vec{A}_1 + \vec{A}_3)\big|_{z=0} = \vec{n} \times \vec{A}_2\big|_{z=0}$

$$\vec{n} \times [\vec{A}_{10} e^{i(k_{1x}x + k_{1y}y - \omega t)} + \vec{A}_{30} e^{i(k_{3x}x + k_{3y}y - \omega_3 t)}] = \vec{n} \times \vec{A}_{20} e^{i(k_{2x}x + k_{2y}y - \omega_2 t)}$$

$$\begin{cases} \omega = \omega_2 = \omega_3 \\ k_{1x} = k_{2x} = k_{3x} = k_1 \sin\theta_1 = \sqrt{\mu_1\epsilon_1}\omega \sin\theta_1 \\ k_{1y} = k_{2y} = k_{3y} = 0 \\ \vec{n} \times (\vec{A}_{10} + \vec{A}_{30}) = \vec{n} \times \vec{A}_{20} \end{cases}$$



- 反射波：

$$\omega_3=\omega,\qquad \mathbf{k_3}=\mathbf{k_1}$$
$$\vec{\mathbf{k}}_3=\mathbf{k}_3\sin\theta_3\vec{\mathbf{e}}_1-\mathbf{k}_3\cos\theta_3\vec{\mathbf{e}}_3$$
$$\mathbf{k_{3x}}=\mathbf{k_{1x}}\quad\rightarrow\quad \mathbf{k}_3\sin\theta_3=\mathbf{k}_1\sin\theta_1\quad\rightarrow\quad \theta_3=\theta_1\quad\text{反射定律}$$

- 透射波：

$$\omega_2=\omega,\qquad \mathbf{k_{2Rx}}=\sqrt{\mu_1\epsilon_1}\omega\sin\theta_1,\qquad \mathbf{k_{2Ix}}=0,\qquad \mathbf{k_{2Ry}}=\mathbf{k_{2Iy}}=0$$
$$\mathbf{k}_{2R}^2-\mathbf{k}_{2I}^2=\mu_2\epsilon_2\omega_2^2,\qquad \vec{\mathbf{k}}_{2R}\cdot\vec{\mathbf{k}}_{2I}=\frac{1}{2}\omega_2\mu_2\gamma_2$$
$$\mathbf{k}_{2Rz}^2-\mathbf{k}_{2Iz}^2=\mathbf{k}_{2R}^2-\mathbf{k}_{2Rx}^2-\mathbf{k}_{2Ry}^2-\mathbf{k}_{2I}^2+\mathbf{k}_{2Ix}^2+\mathbf{k}_{2Iy}^2$$
$$=\mu_2\epsilon_2\omega^2-\mu_1\epsilon_1\omega^2\sin^2\theta_1$$
$$\mathbf{k}_{2Rz}\mathbf{k}_{2Iz}=\vec{\mathbf{k}}_{2R}\cdot\vec{\mathbf{k}}_{2I}-\mathbf{k}_{2Rx}\mathbf{k}_{2Ix}-\mathbf{k}_{2Ry}\mathbf{k}_{2Iy}=\frac{1}{2}\omega\mu_2\gamma_2$$



- 透射波：

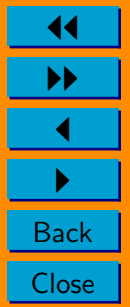
$$\omega_2 = \omega, \qquad \mathbf{k_{2Rx}} = \sqrt{\mu_1\epsilon_1}\omega\sin\theta_1, \qquad \mathbf{k_{2Ix}} = \mathbf{0}, \qquad \mathbf{k_{2Ry}} = \mathbf{k_{2Iy}} = \mathbf{0}$$

$$\mathbf{k_{2Rz}^2} - \mathbf{k_{2Iz}^2} = \mu_2\epsilon_2\omega^2 - \mu_1\epsilon_1\omega^2\sin^2\theta_1 \qquad \mathbf{k_{2Rz}k_{2Iz}} = \frac{1}{2}\omega\mu_2\gamma_2$$

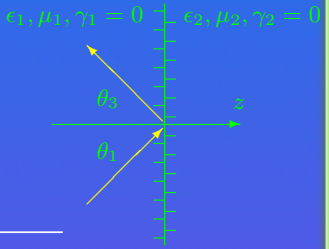
寻求实的、正的 $\mathbf{k_{2Rz}}$ 解

$$\mathbf{k_{2Rz}} = \frac{1}{\sqrt{2}}\sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1)\omega^2 + \sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1)^2\omega^4 + \mu_2^2\gamma_2^2\omega^2}}$$

$$\mathbf{k_{2Iz}} = \frac{\omega\mu_2\gamma_2}{\sqrt{2}\sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1)\omega^2 + \sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1)^2\omega^4 + \mu_2^2\gamma_2^2\omega^2}}}$$



电磁波在界面上的反射和折射： 反射透射波的波矢~绝缘透射介质



$\gamma_2 = 0$ 折射率: $\mathbf{n} \equiv \sqrt{\mu_r \epsilon_r} = \mathbf{c}\sqrt{\mu\epsilon}$

$$\mathbf{k}_2 = \sqrt{\mu_2\epsilon_2}\omega = \frac{\mathbf{n}_2}{\mathbf{c}}\omega \qquad \frac{\mathbf{k}_2}{\mathbf{k}_1} = \frac{\mathbf{n}_2}{\mathbf{n}_1}$$

$$\mathbf{k}_{2Rz} = \frac{\omega}{\sqrt{2}}\sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1) + \|\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1\|}$$

$$\mathbf{k}_{2Iz} = \frac{\mu_2\gamma_2}{\sqrt{2}\sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1) + \|\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1\|}}$$

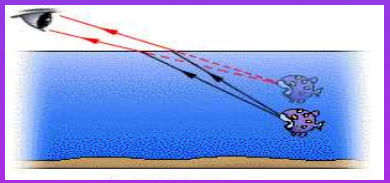
$\mu_2\epsilon_2 \geq \mu_1\epsilon_1 \rightarrow \mathbf{n}_2 \geq \mathbf{n}_1$, 光疏 $\rightarrow$ 光密

$$\mathbf{k}_{2Rz} = \omega\sqrt{\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1} \qquad \mathbf{k}_{2Iz} = 0 \quad \text{无衰减}$$

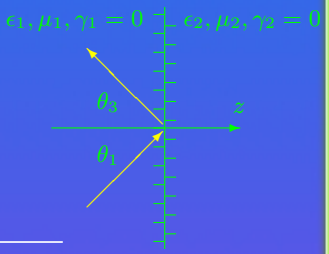
$$\vec{\mathbf{k}}_2 = \mathbf{k}_2\sin\theta_2\vec{\mathbf{e}}_1 + \mathbf{k}_2\cos\theta_2\vec{\mathbf{e}}_3 = \omega\sqrt{\mu_2\epsilon_2}(\sin\theta_2\vec{\mathbf{e}}_1 + \cos\theta_2\vec{\mathbf{e}}_3)$$

$\rightarrow \omega\sqrt{\mu_2\epsilon_2}\sin\theta_2 = \omega\sqrt{\mu_1\epsilon_1}\sin\theta_1 \rightarrow \mathbf{n}_2\sin\theta_2 = \mathbf{n}_1\sin\theta_1$ 折射定律

$$\sin\theta_2 = \frac{\mathbf{n}_1}{\mathbf{n}_2}\sin\theta_1 \leq 1$$



电磁波在界面上的反射和折射： 反射透射波的波矢~绝缘透射介质



$\gamma_2 = 0$ 折射率: $\mathbf{n} \equiv \sqrt{\mu_r \epsilon_r} = \mathbf{c}\sqrt{\mu\epsilon}$

$\mathbf{k_2} = \sqrt{\mu_2\epsilon_2}\omega = \dfrac{\mathbf{n_2}}{\mathbf{c}}\omega \qquad \dfrac{\mathbf{k_2}}{\mathbf{k_1}} = \dfrac{\mathbf{n_2}}{\mathbf{n_1}}$

$$\mathbf{k_{2Rz}} = \frac{\omega}{\sqrt{2}}\sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1) + \|\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1\|}$$

$$\mathbf{k_{2Iz}} = \frac{\mu_2\gamma_2}{\sqrt{2}\sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1) + \|\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1\|}}$$

$\mu_2\epsilon_2 < \mu_1\epsilon_1$ $\sin\theta_1 \le \sqrt{\dfrac{\mu_2\epsilon_2}{\mu_1\epsilon_1}} = \dfrac{\mathbf{n_2}}{\mathbf{n_1}}$, 光密 $\to$ 光疏

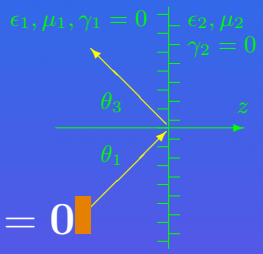
$\mathbf{k_{2Rz}} = \omega\sqrt{\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1}$ $\mathbf{k_{2Iz}} = 0$ 无衰减

$\vec{\mathbf{k}}_2 = \mathbf{k_2}\sin\theta_2\vec{\mathbf{e}}_1 + \mathbf{k_2}\cos\theta_2\vec{\mathbf{e}}_3 = \omega\sqrt{\mu_2\epsilon_2}(\sin\theta_2\vec{\mathbf{e}}_1 + \cos\theta_2\vec{\mathbf{e}}_3)$

$\to$ $\omega\sqrt{\mu_2\epsilon_2}\sin\theta_2 = \omega\sqrt{\mu_1\epsilon_1}\sin\theta_1 \to \mathbf{n_2}\sin\theta_2 = \mathbf{n_1}\sin\theta_1$ 折射定律

$$\sin\theta_2 = \frac{\mathbf{n_1}}{\mathbf{n_2}}\sin\theta_1 \le 1$$

电磁波在界面上的反射和折射：反射透射波的波矢~绝缘透射介质



$$\mu_2\epsilon_2 < \mu_1\epsilon_1 \qquad \sin\theta_1 > \sqrt{\frac{\mu_2\epsilon_2}{\mu_1\epsilon_1}} = \frac{n_2}{n_1},\ \text{光密} \to \text{光疏}$$

$$k_{2Rz} = \frac{\omega}{\sqrt{2}}\sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1) + \|\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1\|} = 0$$

$$\sqrt{2}k_{2Iz} = \frac{\mu_2\gamma_2}{\sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1) + \|\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1\|}} = \frac{0}{0}$$

$$= \frac{\omega\mu_2\gamma_2}{\sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1)\omega^2 + \sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1)^2\omega^4 + \mu_2^2\gamma_2^2\omega^2}}}$$

$$= \frac{\omega\mu_2\gamma_2}{\sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1\sin^2\theta_1)\omega^2 + (\mu_1\epsilon_1\sin^2\theta_1 - \mu_2\epsilon_2)\omega^2\left[1 + \frac{\mu_2^2\gamma_2^2\omega^2}{2(\mu_1\epsilon_1\sin^2\theta_1 - \mu_2\epsilon_2)^2\omega^4}\right]}}\Bigg|_{\gamma_2\to 0}$$

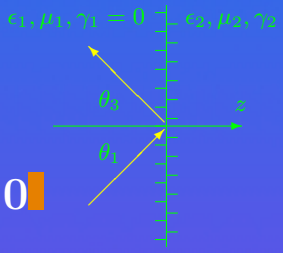
$$= \sqrt{2}\omega\sqrt{\mu_1\epsilon_1\sin^2\theta_1 - \mu_2\epsilon_2} \qquad \vec{k}_{2I} = \omega\sqrt{\mu_1\epsilon_1\sin^2\theta_1 - \mu_2\epsilon_2}\,\vec{e}_3$$

$$\vec{k}_{2R} = \sqrt{\mu_1\epsilon_1}\omega\sin\theta_1\vec{e}_1 \qquad v_{2\phi} = v_{2\phi,x} = \frac{\omega}{k_R} = \frac{1}{\sqrt{\mu_1\epsilon_1}\sin\theta_1} = \frac{v_1}{\sin\theta_1}$$

$$\vec{A}_2 = \vec{A}_{20}e^{-\omega z\sqrt{\mu_1\epsilon_1\sin\theta_1 - \mu_2\epsilon_2}}e^{i(\sqrt{\mu_1\epsilon_1}\omega\sin\theta_1 x - \omega t)}$$

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$\gamma \neq 0$ $\quad \vec{A}_2 = \vec{A}_{20} e^{-k_{2Iz}z} e^{i(k_{2Rx}x + k_{2Rz}z - \omega t)}$

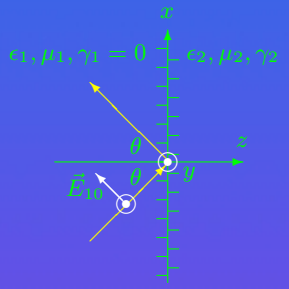
$k_{2Rx} = \sqrt{\mu_1\epsilon_1}\omega \sin\theta_1, \quad k_{2Ix} = 0, \quad k_{2Ry} = k_{2Iy} = 0$

$$k_{2Rz} = \frac{1}{\sqrt{2}}\sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1 \sin^2\theta_1)\omega^2 + \sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1 \sin^2\theta_1)^2\omega^4 + \mu_2^2\gamma_2^2\omega^2}}$$

$$k_{2Iz} = \frac{\omega\mu_2\gamma_2}{\sqrt{2}\sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1 \sin^2\theta_1)\omega^2 + \sqrt{(\mu_2\epsilon_2 - \mu_1\epsilon_1 \sin^2\theta_1)^2\omega^4 + \mu_2^2\gamma_2^2\omega^2}}}$$

透射波等相面沿$\vec{k}_{2R} = k_{2Rx}\vec{e}_1 + k_{2Rz}\vec{e}_3$方向传播,频率$\omega$,速率$v = \frac{\omega}{k_{2R}} = \frac{\omega}{\sqrt{k_{2Rx}^2 + k_{2RZ}^2}}$. 透射波有衰减,方向在z方向.

电磁波在界面上的反射和折射： 反射透射波的振幅~入射波电场位于入射面



$$\begin{cases} \vec{k}_3 \cdot \vec{E}_{30} = 0 \\ \vec{k}_2 \cdot \vec{E}_{20} = 0 \end{cases} \rightarrow \begin{cases} k_{3x}E_{30x} + k_{3z}E_{30z} = 0 \\ k_{2x}E_{20x} + k_{2z}E_{20z} = 0 \end{cases}$$

$$\rightarrow \begin{cases} E_{30z} = -\frac{k_{3x}}{k_{3z}}E_{30x} \\ E_{20z} = -\frac{k_{2x}}{k_{2z}}E_{20x} \end{cases} \qquad \begin{cases} \vec{B}_{30} = \frac{\vec{k}_3}{\omega} \times \vec{E}_{30} \\ \vec{B}_{20} = \frac{k_2}{\omega} \times \vec{E}_{20} \end{cases}$$

$$\vec{E}_{10} = E_{10}(\vec{e}_1 \cos\theta_1 - \vec{e}_3 \sin\theta_1) \qquad \vec{B}_{10} = \frac{\vec{k}_1}{\omega} \times \vec{E}_{10} = \frac{k_1}{\omega}E_{10}\vec{e}_y$$

$$\vec{E}_{10t} + \vec{E}_{30t} = \vec{E}_{20t} \qquad \vec{H}_{10t} + \vec{H}_{30t} = \vec{H}_{20t}$$

$$\begin{cases} E_{30y} = E_{20y} & \clubsuit \\ \frac{1}{\mu_1}B_{30x} = \frac{1}{\mu_2}B_{20x} & \diamondsuit \end{cases} \qquad \begin{cases} E_{10}\cos\theta_1 + E_{30x} = E_{20x} & \spadesuit \\ \frac{1}{\mu_1}\frac{k_1}{\omega}E_{10} + \frac{1}{\mu_1}B_{30y} = \frac{1}{\mu_2}B_{20y} & \heartsuit \end{cases}$$

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电磁波在界面上的反射和折射： 反射透射波的振幅~入射波电场位于入射面

$$\begin{cases} \mathbf{E_{30y}} = \mathbf{E_{20y}} & \clubsuit \\ \frac{1}{\mu_1}\mathbf{B_{30x}} = \frac{1}{\mu_2}\mathbf{B_{20x}} & \diamondsuit \end{cases} \qquad \begin{cases} \mathbf{E_{10}}\cos\theta_1 + \mathbf{E_{30x}} = \mathbf{E_{20x}} & \spadesuit \\ \frac{1}{\mu_1}\frac{\mathbf{k_1}}{\omega}\mathbf{E_{10}} + \frac{1}{\mu_1}\mathbf{B_{30y}} = \frac{1}{\mu_2}\mathbf{B_{20y}} & \heartsuit \end{cases}$$

$$\diamondsuit \rightarrow \frac{1}{\mu_1}(\frac{\vec{\mathbf{k}}_3}{\omega} \times \vec{\mathbf{E}}_{30})_{\mathbf{x}} = \frac{1}{\mu_2}(\frac{\vec{\mathbf{k}}_2}{\omega} \times \vec{\mathbf{E}}_{20})_{\mathbf{x}}$$

$$\rightarrow \frac{1}{\mu_1}(\mathbf{k_{3y}E_{30z}} - \mathbf{k_{3z}E_{30y}}) = \frac{1}{\mu_2}(\mathbf{k_{2y}E_{20z}} - \mathbf{k_{2z}E_{20y}})$$

$$\rightarrow \frac{1}{\mu_1}\mathbf{k_{3z}E_{30y}} = \frac{1}{\mu_2}\mathbf{k_{2z}E_{20y}} \overset{\clubsuit}{\rightarrow} \mathbf{E_{30y}} = \mathbf{E_{20y}} = 0 \overset{\diamondsuit}{\rightarrow} \mathbf{B_{30x}} = \mathbf{B_{20x}} = 0$$

$$\vec{\mathbf{e}}_{2\parallel} = \frac{\mathbf{k_{2z}}}{\mathbf{k_2}}\vec{\mathbf{e}}_1 - \frac{\mathbf{k_{2x}}}{\mathbf{k_2}}\vec{\mathbf{e}}_3 \perp \vec{\mathbf{k}}_2 \qquad \vec{\mathbf{e}}_{3\parallel} = \frac{\mathbf{k_{3z}}}{\mathbf{k_3}}\vec{\mathbf{e}}_1 - \frac{\mathbf{k_{3x}}}{\mathbf{k_3}}\vec{\mathbf{e}}_3 = -\cos\theta_1\vec{\mathbf{e}}_1 - \sin\theta_1\vec{\mathbf{e}}_3 \perp \vec{\mathbf{k}}_3$$

$$\vec{\mathbf{E}}_{20} = \mathbf{E_{20x}}\vec{\mathbf{e}}_1 + \mathbf{E_{20z}}\vec{\mathbf{e}}_3 = \mathbf{E_{20x}}\vec{\mathbf{e}}_1 - \frac{\mathbf{k_{2x}}}{\mathbf{k_{2z}}}\vec{\mathbf{e}}_3\mathbf{E_{20x}} = \vec{\mathbf{e}}_{2\parallel}\frac{\mathbf{k_2}}{\mathbf{k_{2z}}}\mathbf{E_{20x}}$$

$$\vec{\mathbf{E}}_{30} = \mathbf{E_{30x}}\vec{\mathbf{e}}_1 + \mathbf{E_{30z}}\vec{\mathbf{e}}_3 = \mathbf{E_{30x}}\vec{\mathbf{e}}_1 - \frac{\mathbf{k_{3x}}}{\mathbf{k_{3z}}}\vec{\mathbf{e}}_3\mathbf{E_{30x}} = \vec{\mathbf{e}}_{3\parallel}\frac{\mathbf{k_3}}{\mathbf{k_{3z}}}\mathbf{E_{30x}} = -\frac{\vec{\mathbf{e}}_{3\parallel}}{\cos\theta_1}\mathbf{E_{30x}}$$

电磁波在界面上的反射和折射： 反射透射波的振幅~入射波电场位于入射面

$$\begin{cases} \mathbf{E_{30y}} = \mathbf{E_{20y}} & \clubsuit \\ \frac{1}{\mu_1}\mathbf{B_{30x}} = \frac{1}{\mu_2}\mathbf{B_{20x}} & \diamondsuit \end{cases} \qquad \begin{cases} \mathbf{E_{10}}\cos\theta_1 + \mathbf{E_{30x}} = \mathbf{E_{20x}} & \spadesuit \\ \frac{1}{\mu_1}\frac{\mathbf{k_1}}{\omega}\mathbf{E_{10}} + \frac{1}{\mu_1}\mathbf{B_{30y}} = \frac{1}{\mu_2}\mathbf{B_{20y}} & \heartsuit \end{cases}$$

$$\heartsuit \rightarrow \frac{1}{\mu_2}(\frac{\vec{\mathbf{k}}_2}{\omega} \times \vec{\mathbf{E}}_{20})_y - \frac{1}{\mu_1}(\frac{\vec{\mathbf{k}}_3}{\omega} \times \vec{\mathbf{E}}_{30})_y = \frac{1}{\mu_1}\frac{\mathbf{k_1}}{\omega}\mathbf{E_{10}}$$

$$\rightarrow \frac{1}{\mu_2}(\mathbf{k_{2z}}\mathbf{E_{20x}} - \mathbf{k_{2x}}\mathbf{E_{20z}}) - \frac{1}{\mu_1}(\mathbf{k_{3z}}\mathbf{E_{30x}} - \mathbf{k_{3x}}\mathbf{E_{30z}}) = \frac{\mathbf{k_1}}{\mu_1}\mathbf{E_{10}}$$

$$\rightarrow \frac{1}{\mu_2}(\mathbf{k_{2z}} + \frac{\mathbf{k_{2x}^2}}{\mathbf{k_{2z}}})\mathbf{E_{20x}} - \frac{1}{\mu_1}(\mathbf{k_{3z}} + \frac{\mathbf{k_{3x}^2}}{\mathbf{k_{3z}}})\mathbf{E_{30x}} = \frac{1}{\mu_1}\mathbf{k_1}\mathbf{E_{10}}$$

$$\rightarrow \frac{1}{\mu_2}\frac{\mathbf{k_2^2}}{\mathbf{k_{2z}}}\mathbf{E_{20x}} - \frac{1}{\mu_1}\frac{\mathbf{k_3^2}}{\mathbf{k_{3z}}}\mathbf{E_{30x}} = \frac{1}{\mu_1}\mathbf{k_1}\mathbf{E_{10}}$$

$$+\spadesuit \rightarrow \frac{\mu_1}{\mu_2}\frac{\mathbf{k_2^2}}{\mathbf{k_1}\mathbf{k_{2z}}}\mathbf{E_{20x}} + \frac{1}{\cos\theta_1}\mathbf{E_{30x}} = \mathbf{E_{10}} \qquad \mathbf{E_{20x}} - \mathbf{E_{30x}} = \mathbf{E_{10}}\cos\theta_1$$

$$\rightarrow \mathbf{E_{20x}} = \frac{2\cos\theta_1}{1 + \frac{\mu_1\mathbf{k_2^2}\cos\theta_1}{\mu_2\mathbf{k_1}\mathbf{k_{2z}}}}\mathbf{E_{10}} \qquad \mathbf{E_{30x}} = \frac{1 - \frac{\mu_1\mathbf{k_2^2}\cos\theta_1}{\mu_2\mathbf{k_1}\mathbf{k_{2z}}}}{1 + \frac{\mu_1\mathbf{k_2^2}\cos\theta_1}{\mu_2\mathbf{k_1}\mathbf{k_{2z}}}}\cos\theta_1\mathbf{E_{10}}$$



电磁波在界面上的反射和折射： 反射透射波的振幅~入射波电场位于入射面

$$\rightarrow \mathbf{E_{20x}} = \frac{2\cos\theta_1}{1+\frac{\mu_1 k_2^2 \cos\theta_1}{\mu_2 k_1 k_{2z}}}\mathbf{E_{10}} \qquad \mathbf{E_{30x}} = \frac{1-\frac{\mu_1 k_2^2 \cos\theta_1}{\mu_2 k_1 k_{2z}}}{1+\frac{\mu_1 k_2^2 \cos\theta_1}{\mu_2 k_1 k_{2z}}}\cos\theta_1 \mathbf{E_{10}}$$

$$\vec{\mathbf{E}}_{20\parallel} = \frac{2\cos\theta_1}{\frac{k_{2z}}{k_2}+\frac{k_2\mu_1}{k_1\mu_2}\cos\theta_1}\mathbf{E_{10\parallel}}\vec{e}_{2\parallel} \qquad \vec{\mathbf{E}}_{30\parallel} = \frac{-\frac{k_{2z}}{k_2}+\frac{k_2\mu_1}{k_1\mu_2}\cos\theta_1}{\frac{k_{2z}}{k_2}+\frac{k_2\mu_1}{k_1\mu_2}\cos\theta_1}\mathbf{E_{10\parallel}}\vec{e}_{3\parallel}$$

$$\gamma_2 = 0, \mu_2 = \mu_1 \rightarrow k_{2z} = k_2\cos\theta_2;\ \ k_{2x} = k_2\sin\theta_2$$

$$\rightarrow \vec{e}_{2\parallel} = \cos\theta_2\vec{e}_1 - \sin\theta_2\vec{e}_3;\ \ k_2/k_1 = n_2/n_1 = \sin\theta_1/\sin\theta_2$$

$$\vec{\mathbf{E}}_{20\parallel} = \frac{2\cos\theta_1}{\cos\theta_2+\frac{\sin\theta_1}{\sin\theta_2}\cos\theta_1}\mathbf{E_{10\parallel}}\vec{e}_{2\parallel} = \frac{2\cos\theta_1\sin\theta_2}{\frac{1}{2}\sin 2\theta_2+\frac{1}{2}\sin 2\theta_1}\mathbf{E_{10\parallel}}\vec{e}_{2\parallel}$$

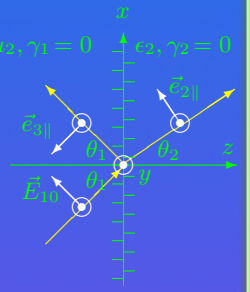
$$= \frac{2\cos\theta_1\sin\theta_2}{\sin(\theta_1+\theta_2)\cos(\theta_1-\theta_2)}\mathbf{E_{10\parallel}}\vec{e}_{2\parallel}$$

$$\vec{\mathbf{E}}_{30\parallel} = \frac{-\sin\theta_2\cos\theta_2+\sin\theta_1\cos\theta_1}{\sin\theta_2\cos\theta_2+\sin\theta_1\cos\theta_1}\mathbf{E_{10\parallel}}\vec{e}_{3\parallel} = \frac{-\sin 2\theta_2+\sin 2\theta_1}{\sin 2\theta_2+\sin 2\theta_1}\mathbf{E_{10\parallel}}\vec{e}_{3\parallel}$$

$$= \frac{\tan(\theta_1-\theta_2)}{\tan(\theta_1+\theta_2)}\mathbf{E_{10\parallel}}\vec{e}_{3\parallel}$$



电磁波在界面上的反射和折射：反射透射波的振幅~入射波电场位于入射面



$\gamma_2=0,\ \mu_2=\mu_1$ $\qquad$ $\mathbf{k_2/k_1=n_2/n_1}=\sin\theta_1/\sin\theta_2$

$\vec{\mathbf{e}}_{2\parallel}=\cos\theta_2\vec{\mathbf{e}}_1-\sin\theta_2\vec{\mathbf{e}}_3$ $\qquad$ $\vec{\mathbf{e}}_{3\parallel}=-\cos\theta_1\vec{\mathbf{e}}_1-\sin\theta_1\vec{\mathbf{e}}_3$

$$\vec{\mathbf{E}}_{30\parallel}=\frac{\tan(\theta_1-\theta_2)}{\tan(\theta_1+\theta_2)}\mathbf{E}_{10\parallel}\vec{\mathbf{e}}_{3\parallel}=\mathbf{E}_{10\parallel}\vec{\mathbf{e}}_{3\parallel}\times\begin{cases}\frac{\mathbf{n_2-n_1}}{\mathbf{n_2+n_1}} & \theta_1\to 0\\ -1 & \theta_1\to\frac{\pi}{2}\end{cases}$$

这时$\theta_2<\frac{\pi}{2}$入射反射波完全抵销

$n_1=n_2\sin\theta_2\le n_2$ 若要求θ_2为实⇑则只允许光疏到光密的情形

光疏→光密$(\mathbf{n_2>n_1})\Rightarrow$ $\vec{\mathbf{E}}_{30\parallel}$ 平行于 $\theta_1\to 0$ $\vec{\mathbf{e}}_{3\parallel}\Rightarrow$ 无半波损失

反平行于 $\theta_1\to\frac{\pi}{2}$ $\vec{\mathbf{e}}_{3\parallel}\Rightarrow$ 半波损失

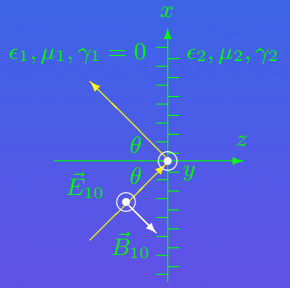
光密→光疏$(\mathbf{n_1>n_2})\Rightarrow$ $\theta_1\to 0$有半波损；θ_1虽在θ_2的实数域无法取$\frac{\pi}{2}$，但在复数域可取，结果有半波损

$$\vec{\mathbf{E}}_{20\parallel}=\frac{\mathbf{2}\cos\theta_1\sin\theta_2}{\sin(\theta_1+\theta_2)\cos(\theta_1-\theta_2)}\mathbf{E}_{10\parallel}\vec{\mathbf{e}}_{2\parallel}=\mathbf{E}_{10\parallel}\vec{\mathbf{e}}_{2\parallel}\times\begin{cases}\frac{\mathbf{2n_1}}{\mathbf{n_2+n_1}} & \theta_1\to 0\\ 0 & \theta_1\to\frac{\pi}{2}\end{cases}$$

$\Rightarrow$ $\vec{\mathbf{E}}_{20\parallel}$平行于$\vec{\mathbf{e}}_{2\parallel}$ $\Rightarrow$ 无半波损失

这时入射波与反射波完全相消⇑因而不存在透射波！

电磁波在界面上的反射和折射： 反射透射波的振幅~入射波电场垂直入射面



$$\begin{cases} \vec{k}_3 \cdot \vec{E}_{30} = 0 \\ \vec{k}_2 \cdot \vec{E}_{20} = 0 \end{cases} \longrightarrow \begin{cases} k_{3x}E_{30x} + k_{3z}E_{30z} = 0 \\ k_{2x}E_{20x} + k_{2z}E_{20z} = 0 \end{cases}$$

$$\rightarrow \begin{cases} E_{30z} = -\frac{k_{3x}}{k_{3z}}E_{30x} \\ E_{20z} = -\frac{k_{2x}}{k_{2z}}E_{20x} \end{cases} \qquad \begin{cases} \vec{B}_{30} = \frac{\vec{k}_3}{\omega} \times \vec{E}_{30} \\ \vec{B}_{20} = \frac{k_2}{\omega} \times \vec{E}_{20} \end{cases}$$

$$\vec{E}_{10} = E_{10}\vec{e}_2 \qquad \vec{B}_{10} = \frac{\vec{k}_1}{\omega} \times \vec{E}_{10} = \frac{k_1}{\omega}E_{10}(-\vec{e}_1\cos\theta_1 + \vec{e}_3\sin\theta_1)$$

$$\vec{E}_{10t} + \vec{E}_{30t} = \vec{E}_{20t} \qquad \vec{H}_{10t} + \vec{H}_{30t} = \vec{H}_{20t}$$

$$\begin{cases} E_{30x} = E_{20x} & \clubsuit \\ \frac{1}{\mu_1}B_{30y} = \frac{1}{\mu_2}B_{20y} & \diamondsuit \end{cases} \qquad \begin{cases} E_{10y} + E_{30y} = E_{20y} & \spadesuit \\ \frac{1}{\mu_1}(B_{10x} + B_{30x}) = \frac{1}{\mu_2}B_{20x} & \heartsuit \end{cases}$$

电磁波在界面上的反射和折射：反射透射波的振幅~入射波电场垂直入射面

$$\vec{E}_{20\perp}=\frac{2}{1+\frac{\mu_1 k_{2z}}{\mu_2 k_1\cos\theta_1}}E_{10\perp}\vec{e}_2 \qquad \vec{E}_{30\perp}=\frac{1-\frac{\mu_1 k_{2z}}{\mu_2 k_1\cos\theta_1}}{1+\frac{\mu_1 k_{2z}}{\mu_2 k_1\cos\theta_1}}E_{10\perp}\vec{e}_2$$

$$\gamma_2=0,\mu_2=\mu_1\rightarrow k_{2z}=k_2\cos\theta_2=k_1\sin\theta_1\cos\theta_2/\sin\theta_2$$

$$\vec{E}_{30\perp}=\frac{1-\frac{\sin\theta_1\cos\theta_2}{\sin\theta_2\cos\theta_1}}{1+\frac{\sin\theta_1\cos\theta_2}{\sin\theta_2\cos\theta_1}}E_{10\perp}\vec{e}_2=E_{10\perp}\vec{e}_2\times\begin{cases}\frac{n_1-n_2}{n_1+n_2} & \theta_1\to 0\\ -1 & \theta_1\to\frac{\pi}{2}\end{cases}$$

这时$\theta_2<\frac{\pi}{2}$入射反射波完全抵销

光疏→光密$(n_2>n_1)\Rightarrow$ $\vec{E}_{30\perp}$ 反平行于 $\theta_1\to 0$；反平行于 $\theta_1\to\frac{\pi}{2}$ $\vec{e}_2\Rightarrow$ 半波损失 半波损失

光密→光疏$(n_1>n_2)\Rightarrow$ $\theta_1\to 0$无半波损；θ_1虽在θ_2的实数域无法取$\frac{\pi}{2}$，但在复数域可取，结果有半波损

$$\vec{E}_{20\perp}=\frac{2}{1+\frac{\sin\theta_1\cos\theta_2}{\sin\theta_2\cos\theta_1}}E_{10\perp}\vec{e}_2=E_{10\perp}\vec{e}_2\times\begin{cases}\frac{2n_1}{n_1+n_2} & \theta_1\to 0\\ 0 & \theta_1\to\frac{\pi}{2}\end{cases}$$

入射反射抵消，无透射

光疏→光密 $\Rightarrow$ $n_2>n_1$ $\Rightarrow$ $\vec{E}_{20\perp}$ 平行于 $\vec{e}_2$ 无半波损失

电磁波在界面上的反射和折射：反射波的振幅~一般情况

$$\vec{k}_1 = k_1(\sin\theta_1\vec{e}_1 + \cos\theta_1\vec{e}_3) \qquad \vec{k}_3 = k_3(\sin\theta_1\vec{e}_1 - \cos\theta_1\vec{e}_3) \qquad k_1 = k_3$$

$$\vec{E}_{10} = \vec{E}_{10\parallel} + \vec{E}_{10\perp} = E_{10\parallel}(\vec{e}_1\cos\theta_1 - \vec{e}_3\sin\theta_1) + E_{10\perp}\vec{e}_2$$

$$= E_{10\parallel}(\vec{e}_2 \times \frac{\vec{k}_1}{k_1}) + E_{10\perp}\vec{e}_2 \qquad \vec{e}_2 = \frac{\vec{e}_3\times\vec{k}_1}{|\vec{e}_3\times\vec{k}_1|} = \frac{\vec{e}_3\times\vec{k}_3}{|\vec{e}_3\times\vec{k}_3|}$$

$$\vec{E}_{30} = \vec{E}_{30\parallel} + \vec{E}_{30\perp}$$

$$= E_{10\parallel}(\vec{e}_1\cos\theta_1 + \vec{e}_3\sin\theta_1)\frac{\frac{k_{2z}}{k_2} - \frac{\mu_1 k_2}{\mu_2 k_1}\cos\theta_1}{\frac{k_{2z}}{k_2} + \frac{\mu_1 k_2}{\mu_2 k_1}\cos\theta_1} + E_{10\perp}\vec{e}_2\frac{1 - \frac{\mu_1 k_{2z}}{\mu_2 k_1\cos\theta_1}}{1 + \frac{\mu_1 k_{2z}}{\mu_2 k_1\cos\theta_1}}$$

$$= E_{10\parallel}(\vec{e}_2 \times \frac{\vec{k}_3}{k_3})\frac{-\frac{k_{2z}}{k_2} + \frac{\mu_1 k_2}{\mu_2 k_1}\cos\theta_1}{\frac{k_{2z}}{k_2} + \frac{\mu_1 k_2}{\mu_2 k_1}\cos\theta_1} + E_{10\perp}\vec{e}_2\frac{1 - \frac{\mu_1 k_{2z}}{\mu_2 k_1\cos\theta_1}}{1 + \frac{\mu_1 k_{2z}}{\mu_2 k_1\cos\theta_1}}$$

在全反射时，k_{2z}是纯虚数，k_1, k_2是实数；　⇑系数是一个纯相角　⇑系数是一个纯相角

$$\overset{\gamma_2=0,\ \mu_2=\mu_1}{=\!=\!=\!=\!=\!=\!=} E_{10\parallel}(\vec{e}_2 \times \frac{\vec{k}_3}{k_3})\frac{\tan(\theta_1-\theta_2)}{\tan(\theta_1+\theta_2)} + E_{10\perp}\vec{e}_2\frac{\sin(\theta_2-\theta_1)}{\sin(\theta_1+\theta_2)}$$

$$\overset{\theta_1=0}{=\!=\!=} \frac{n_2 - n_1}{n_2 + n_1}[E_{10\parallel}(\vec{e}_2 \times \frac{\vec{k}_3}{k_3}) - E_{10\perp}\vec{e}_2]$$

右旋 ⇔ 左旋

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电磁波在界面上的反射和折射：透射波的振幅~一般情况

$$\vec{k}_1 = k_1(\sin\theta_1\vec{e}_1 + \cos\theta_1\vec{e}_3) \qquad \vec{k}_3 = k_3(\sin\theta_1\vec{e}_1 - \cos\theta_1\vec{e}_3) \qquad k_1 = k_3$$

$$\vec{E}_{10} = \vec{E}_{10\parallel} + \vec{E}_{10\perp} = E_{10\parallel}(\vec{e}_1\cos\theta_1 - \vec{e}_3\sin\theta_1) + E_{10\perp}\vec{e}_2$$

$$= E_{10\parallel}(\vec{e}_2 \times \frac{\vec{k}_1}{k_1}) + E_{10\perp}\vec{e}_2$$

$$\vec{E}_{20} = \vec{E}_{20\parallel} + \vec{E}_{20\perp}$$

$$= E_{10\parallel}(\vec{e}_1\frac{k_{2z}}{k_2} - \vec{e}_3\frac{k_{2x}}{k_2})\frac{2\cos\theta_1}{\frac{k_{2z}}{k_2} + \frac{\mu_1 k_2}{\mu_2 k_1}\cos\theta_1} + E_{10\perp}\vec{e}_2\frac{2}{1 + \frac{\mu_1 k_{2z}}{\mu_2 k_1\cos\theta_1}}$$

$$= E_{10\parallel}(\vec{e}_2 \times \frac{\vec{k}_2}{k_2})\frac{2\cos\theta_1}{\frac{k_{2z}}{k_2} + \frac{\mu_1 k_2}{\mu_2 k_1}\cos\theta_1} + E_{10\perp}\vec{e}_2\frac{2}{1 + \frac{\mu_1 k_{2z}}{\mu_2 k_1\cos\theta_1}}$$

$$\overset{\gamma_2=0,\ \mu_2=\mu_1}{=\!=\!=\!=\!=\!=} E_{10\parallel}(\vec{e}_2 \times \frac{\vec{k}_2}{k_2})\frac{2\cos\theta_1\sin\theta_2}{\sin(\theta_1+\theta_2)\cos(\theta_1-\theta_2)} + E_{10\perp}\vec{e}_2\frac{2\sin\theta_2\cos\theta_1}{\sin(\theta_1+\theta_2)}$$

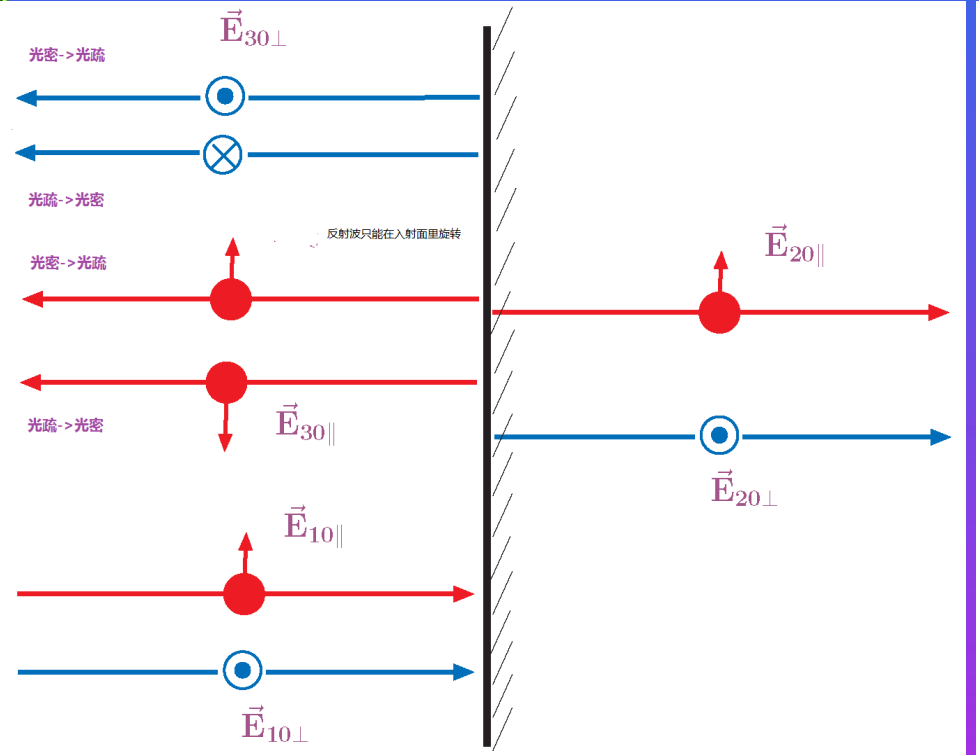
$$\overset{\theta_1=0}{=\!=\!=} \frac{2n_1}{n_2+n_1}[E_{10\parallel}(\vec{e}_2 \times \frac{\vec{k}_2}{k_2}) + E_{10\perp}\vec{e}_2]$$

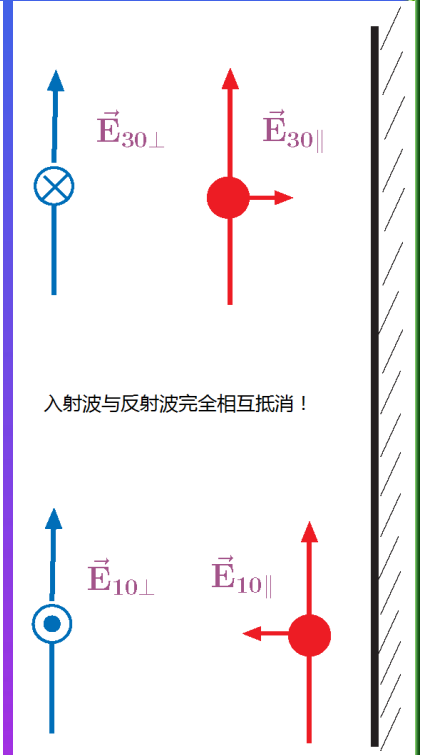
左旋⇔左旋；　右旋⇔右旋

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电磁波在界面上的反射和折射： 垂直入射和掠入射

反射波掠入射及垂直入射光密到光疏时有半波损；反射波其它情形和透射波无半波损！





掠入射时入射反射波完全相消，无透射波的解释：这时观测到的入射波完全不与界面作用！

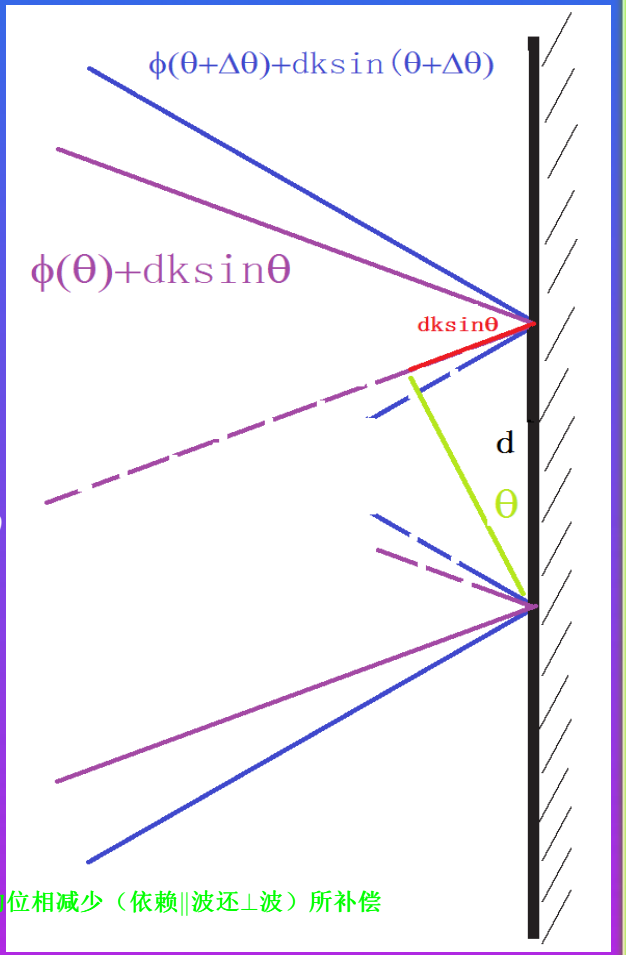
电磁波在界面上的反射和折射： Goos－Hänchen shift)

- 全反射时反射波出现额外的位相$\phi(\theta)$
- 这个位相$\phi(\theta)$依赖入射角θ
- 导致沿界面有位移相异入射角的反射波干涉
- 形成 古斯－汉欣位移
- 设反射波的位相为：$\vec{\mathrm{k}}\cdot\vec{\mathrm{r}}-\omega\mathrm{t}+\phi(\theta)$
- 向上平移入射点d导致位相额外增加：$\mathrm{dk}\sin\theta$
- 入射角θ变化导致位相变化：$\phi(\theta)\Rightarrow\phi(\theta+\Delta\theta)$
- 两种变化结合会产生干涉效应

$$\phi(\theta)+\mathrm{dk}\sin\theta=\phi(\theta+\Delta\theta)+\mathrm{dk}\sin(\theta+\Delta\theta)$$

$$\mathrm{d}=-\frac{1}{\mathrm{k}\cos\theta}\frac{\partial\phi(\theta)}{\partial\theta}$$

空间位移导致的位相增加被入射角导致的位相减少（依赖||波还⊥波）所补偿



电磁波在界面上的反射和折射： Goos－Hänchen shift) 续

$$\vec{S}_{sum} = \frac{1}{2\mu\omega}\mathrm{Re}[(\vec{E}^* + \vec{E}'^*)\times(\vec{k}\times\vec{E} + \vec{k}'\times\vec{E}')]$$ 对两束同频波：$\vec{k}$、$\vec{E}$、$\vec{B}$、$\vec{k}'$、$\vec{E}'$、$\vec{B}'$

$$= \frac{1}{2\mu\omega}\mathrm{Re}[\vec{k}(\vec{E}\cdot\vec{E}^*) - \vec{E}(\vec{k}\cdot\vec{E}^*) + \vec{k}'(\vec{E}'\cdot\vec{E}'^*) - \vec{E}'(\vec{k}'\cdot\vec{E}'^*)$$

$$+\vec{k}'(\vec{E}'\cdot\vec{E}^*) - \vec{E}'(\vec{k}'\cdot\vec{E}^*) + \vec{k}(\vec{E}\cdot\vec{E}'^*) - \vec{E}(\vec{k}\cdot\vec{E}'^*)]$$

$$= \frac{1}{2\mu\omega}\mathrm{Re}\{\vec{k}(\vec{E}_0\cdot\vec{E}_0^*) - \vec{E}_0(\vec{k}\cdot\vec{E}_0^*) + \vec{k}'(\vec{E}_0'\cdot\vec{E}_0'^*) - \vec{E}_0'(\vec{k}'\cdot\vec{E}_0'^*)$$ 两束能流叠加

$$+[\vec{k}'(\vec{E}_0'\cdot\vec{E}_0^*) - \vec{E}_0'(\vec{k}'\cdot\vec{E}_0^*)]e^{i(\vec{k}'-\vec{k})\vec{r}} + [\vec{k}(\vec{E}_0\cdot\vec{E}_0'^*) - \vec{E}_0(\vec{k}\cdot\vec{E}_0'^*)]e^{i(\vec{k}-\vec{k}')\vec{r}}\}$$ 干涉项

入射点r取为零。全反射时反射能流干涉项在入射点出现额外位相，在位移后反射点被抵消！其各项模和入射能流对应项一样。

$$\vec{S}_{3sum入射点} = \frac{1}{2\mu\omega}\mathrm{Re}\{\vec{k}_3(\vec{E}_{10}\cdot\vec{E}_{10}^*) + \vec{k}_3'(\vec{E}_{10}'\cdot\vec{E}_{10}'^*)$$

$$+[\vec{k}_3'(\vec{E}_{30}'\cdot\vec{E}_{30}^*) + \vec{k}_3(\vec{E}_{30}\cdot\vec{E}_{30}'^*)] - [\vec{E}_{30}'(\vec{k}_3'\cdot\vec{E}_{30}^*) + \vec{E}_{30}(\vec{k}_3\cdot\vec{E}_{30}'^*)]\}$$

$$\vec{S}_{3sum位移后反射点} = \frac{1}{2\mu\omega}\mathrm{Re}\{\vec{k}_3(\vec{E}_{10}\cdot\vec{E}_{10}^*) + \vec{k}_3'(\vec{E}_{10}'\cdot\vec{E}_{10}'^*)$$

$$+[\vec{k}_3'(\vec{E}_{30}'\cdot\vec{E}_{30}^*) + \vec{k}_3(\vec{E}_{30}\cdot\vec{E}_{30}'^*)] - [\vec{E}_{30}'(\vec{k}_3'\cdot\vec{E}_{30}^*) + \vec{E}_{30}(\vec{k}_3\cdot\vec{E}_{30}'^*)]\}$$

$$\Uparrow + \frac{\vec{k}_3'\cdot\vec{k}_3}{k_1^2}[\vec{k}_3'(\vec{E}_{10\parallel}'\cdot\vec{E}_{10\parallel}^*) + \vec{k}_3(\vec{E}_{10\parallel}\cdot\vec{E}_{10\parallel}'^*)] + [\vec{k}_3'(\vec{E}_{10\perp}'\cdot\vec{E}_{10\perp}^*) + \vec{k}_3(\vec{E}_{10\perp}\cdot\vec{E}_{10\perp}'^*)]$$

$-\vec{e}_2\cdot(\frac{\vec{k}_3}{k_1}\times\vec{k}_3')[\vec{E}_{30}'E_{30\parallel}^* - \vec{E}_{30}E_{30\parallel}'^*]$ ∥波和⊥波各自的平移距离并不相同，干涉项中的∥和⊥交叉项会有不相消的位相！ ⇑



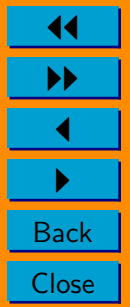
$$\vec{\bar{S}} = \frac{1}{2\mu\omega}\mathrm{Re}[\vec{k}(\vec{E}\cdot\vec{E}^*) - \vec{E}(\vec{k}\cdot\vec{E}^*)]$$

$$\vec{\bar{S}}_1 = \frac{1}{2\mu_1\omega}\mathrm{Re}[\vec{k}_1(\vec{E}_{10}\cdot\vec{E}^*_{10})] = \frac{1}{2\mu_1\omega}\vec{k}_1(\vec{E}_{10\parallel}\cdot\vec{E}^*_{10\parallel} + \vec{E}_{10\perp}\cdot\vec{E}^*_{10\perp})$$
$$= \vec{\bar{S}}_{1\parallel} + \vec{\bar{S}}_{1\perp}$$

$$\vec{k}_1 = \vec{k}_1^* \qquad \vec{k}_1\cdot\vec{E}_{10} = 0$$

$$\vec{\bar{S}}_{1\parallel} = \frac{\vec{k}_1}{2\mu_1\omega}(\vec{E}_{10\parallel}\cdot\vec{E}^*_{10\parallel}) = \frac{\vec{k}_1}{2\mu_1\omega}E_{10\parallel}E^*_{10\parallel}$$

$$\vec{\bar{S}}_{1\perp} = \frac{\vec{k}_1}{2\mu_1\omega}(\vec{E}_{10\perp}\cdot\vec{E}^*_{10\perp}) = \frac{\vec{k}_1}{2\mu_1\omega}E_{10\perp}E^*_{10\perp}$$





$$\vec{S} = \frac{1}{2\mu\omega}\mathrm{Re}[\vec{k}(\vec{E}\cdot\vec{E}^*) - \vec{E}(\vec{k}\cdot\vec{E}^*)]$$

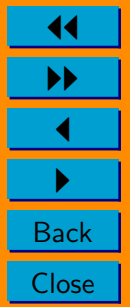
$$\vec{S}_1 = \vec{S}_{1\parallel} + \vec{S}_{1\perp}$$

$$\vec{S}_3 = \frac{1}{2\mu_1\omega}\mathrm{Re}[\vec{k}_3(\vec{E}_{30}\cdot\vec{E}^*_{30})] = \frac{1}{2\mu_1\omega}\vec{k}_3(\vec{E}_{30\parallel}\cdot\vec{E}^*_{30\parallel} + \vec{E}_{30\perp}\cdot\vec{E}^*_{30\perp})$$
$$= \vec{S}_{3\parallel} + \vec{S}_{3\perp}$$

$$\vec{k}_3 = \vec{k}_3^* \qquad \vec{k}_3\cdot\vec{E}_{30} = 0$$

$$\vec{S}_{3\parallel} = \frac{\vec{k}_3}{2\mu_1\omega}(\vec{E}_{30\parallel}\cdot\vec{E}^*_{30\parallel}) = \frac{\vec{k}_3}{2\mu_1\omega}E_{30\parallel}E^*_{30\parallel}$$

$$\vec{S}_{3\perp} = \frac{\vec{k}_3}{2\mu_1\omega}(\vec{E}_{30\perp}\cdot\vec{E}^*_{30\perp}) = \frac{\vec{k}_3}{2\mu_1\omega}E_{30\perp}E^*_{30\perp}$$



$$\overline{\vec{S}} = \frac{1}{2\mu\omega}\mathrm{Re}[\vec{k}(\vec{E}\cdot\vec{E}^*) - \vec{E}(\vec{k}\cdot\vec{E}^*)]$$

$$\overline{\vec{S}_1} = \overline{\vec{S}_{1\parallel}} + \overline{\vec{S}_{1\perp}} \qquad \overline{\vec{S}_3} = \overline{\vec{S}_{3\parallel}} + \overline{\vec{S}_{3\perp}}$$

$$\begin{aligned}\overline{\vec{S}_2} &= \frac{1}{2\mu_2\omega}\mathrm{Re}[\vec{k}_2(\vec{E}_2\cdot\vec{E}_2^*) - \vec{E}_2(\vec{k}_2\cdot\vec{E}_2^*)] \\ &= \frac{1}{2\mu_2\omega}\mathrm{Re}[\vec{k}_2(\vec{E}_{2\parallel}\cdot\vec{E}_{2\parallel}^* + \vec{E}_{2\perp}\cdot\vec{E}_{2\perp}^*) - (\vec{E}_{2\parallel}+\vec{E}_{2\perp})(\vec{k}_2\cdot\vec{E}_{2\parallel}^*)] \\ &= \overline{\vec{S}_{2\parallel}} + \overline{\vec{S}_{2\perp}}\end{aligned}$$

$$\overline{\vec{S}_{2\parallel}} = \frac{1}{2\mu_2\omega}\{\vec{k}_{2R}(\vec{E}_{2\parallel}\cdot\vec{E}_{2\parallel}^*) - \mathrm{Re}(\vec{E}_{2\parallel}+\vec{E}_{2\perp})(\vec{k}_2\cdot\vec{E}_{2\parallel}^*)\} \qquad \text{无下标0}$$

$$\overline{\vec{S}_{2\perp}} = \frac{\vec{k}_{2R}}{2\mu_2\omega}(\vec{E}_{2\perp}\cdot\vec{E}_{2\perp}^*) \qquad \text{无下标0}$$

取界面$(z=0)$上一个小面元$\Delta\vec{S}=\Delta S\vec{e}_3$

$$D_\perp \equiv \frac{\overline{\vec{S}}_{2\perp}\cdot\Delta\vec{S}}{\overline{\vec{S}}_{1\perp}\cdot\Delta\vec{S}}\Big|_{z=0}=\frac{\overline{S}_{2\perp z}}{\overline{S}_{1\perp z}}\Big|_{z=0}=\frac{\frac{k_{2zR}}{2\mu_2\omega}(\vec{E}_{2\perp}\cdot\vec{E}^*_{2\perp})}{\frac{k_{1z}}{2\mu_1\omega}(\vec{E}_{10\perp}\cdot\vec{E}^*_{10\perp})}$$

$$=\frac{\mu_1 k_{2zR}}{\mu_2 k_1\cos\theta_1}\frac{\vec{E}_{20\perp}\cdot\vec{E}^*_{20\perp}}{E_{10\perp}E^*_{10\perp}}e^{-2\vec{k}_{2I}\cdot\vec{r}}\Big|_{z=0}=\frac{\mu_1 k_{2zR}}{\mu_2 k_1\cos\theta_1}\left\|\frac{2}{1+\frac{\mu_1 k_{2z}}{\mu_2 k_1\cos\theta_1}}\right\|^2$$

$$R_\perp \equiv \frac{-\overline{\vec{S}}_{3\perp}\cdot\Delta\vec{S}}{\overline{\vec{S}}_{1\perp}\cdot\Delta\vec{S}}\Big|_{z=0}=-\frac{\overline{S}_{3\perp z}}{\overline{S}_{1\perp z}}\Big|_{z=0}=-\frac{\frac{k_{3z}}{2\mu_1\omega}(\vec{E}_{30\perp}\cdot\vec{E}^*_{30\perp})}{\frac{k_{1z}}{2\mu_1\omega}(\vec{E}_{10\perp}\cdot\vec{E}^*_{10\perp})}$$

$$=\frac{\vec{E}_{30\perp}\cdot\vec{E}^*_{30\perp}}{E_{10\perp}E^*_{10\perp}}=\left\|\frac{1-\frac{\mu_1 k_{2z}}{\mu_2 k_1\cos\theta_1}}{1+\frac{\mu_1 k_{2z}}{\mu_2 k_1\cos\theta_1}}\right\|^2$$



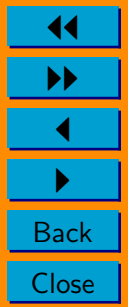
$$\mathbf{R}_\perp + \mathbf{D}_\perp = \frac{\frac{4\mu_1 \mathbf{k}_{2z\mathbf{R}}}{\mu_2 \mathbf{k}_1 \cos\theta_1} + (1 - \frac{\mu_1 \mathbf{k}_{2z}}{\mu_2 \mathbf{k}_1 \cos\theta_1})(1 - \frac{\mu_1 \mathbf{k}^*_{2z}}{\mu_2 \mathbf{k}_1 \cos\theta_1})}{\|1 + \frac{\mu_1 \mathbf{k}_{2z}}{\mu_2 \mathbf{k}_1 \cos\theta_1}\|^2}$$

$$= \frac{\frac{2\mu_1}{\mu_2 \mathbf{k}_1 \cos\theta_1}(\mathbf{k}_{2z} + \mathbf{k}^*_{2z}) + 1 - \frac{\mu_1}{\mu_2 \mathbf{k}_1 \cos\theta_1}(\mathbf{k}_{2z} + \mathbf{k}^*_{2z}) + \frac{\mu_1^2 \mathbf{k}_{2z} \mathbf{k}^*_{2z}}{\mu_2^2 \mathbf{k}_1^2 \cos^2\theta_1}}{\|1 + \frac{\mu_1 \mathbf{k}_{2z}}{\mu_2 \mathbf{k}_1 \cos\theta_1}\|^2}$$

$$= 1$$

$$(\vec{\overline{\mathbf{S}}}_{2\perp} - \vec{\overline{\mathbf{S}}}_{3\perp}) \cdot \Delta\vec{\mathbf{S}} = (\mathbf{D}_\perp + \mathbf{R}_\perp)\vec{\overline{\mathbf{S}}}_{1\perp} \cdot \Delta\vec{\mathbf{S}} = \vec{\overline{\mathbf{S}}}_{1\perp} \cdot \Delta\vec{\mathbf{S}}$$

从入射波流入ΔS的$\perp$电磁能量等于从ΔS流出的 $\perp$的反射波和透射波电磁能量的迭加.



电磁波在界面上的反射和折射：反射透射波的能流

$$\mathbf{D}_{\perp}=\frac{\mu_1\mathbf{k}_{2zR}}{\mu_2\mathbf{k}_1\cos\theta_1}\left\|\frac{2}{1+\frac{\mu_1\mathbf{k}_{2z}}{\mu_2\mathbf{k}_1\cos\theta_1}}\right\|^2 \qquad \mathbf{R}_{\perp}=\left\|\frac{1-\frac{\mu_1\mathbf{k}_{2z}}{\mu_2\mathbf{k}_1\cos\theta_1}}{1+\frac{\mu_1\mathbf{k}_{2z}}{\mu_2\mathbf{k}_1\cos\theta_1}}\right\|^2$$

$$\mathbf{D}_{\parallel}=\frac{4\mu_1\cos\theta_1\mathbf{Re}\left(\frac{\mathbf{k}_2\mathbf{k}_{2z}^*}{\mathbf{k}_2^*}\right)}{\mu_2\mathbf{k}_1\left\|\frac{\mathbf{k}_{2z}}{\mathbf{k}_2}+\frac{\mathbf{k}_2\mu_1}{\mathbf{k}_1\mu_2}\cos\theta_1\right\|^2} \qquad \mathbf{R}_{\parallel}=\left\|\frac{\frac{\mathbf{k}_{2z}}{\mathbf{k}_2}-\frac{\mathbf{k}_2\mu_1}{\mathbf{k}_1\mu_2}\cos\theta_1}{\frac{\mathbf{k}_{2z}}{\mathbf{k}_2}+\frac{\mathbf{k}_2\mu_1}{\mathbf{k}_1\mu_2}\cos\theta_1}\right\|^2$$

例：$\gamma_2=0,\ \mu_1=\mu_2,\ \mathbf{k}_2=\mathbf{k}_1\sin\theta_1/\sin\theta_2,\ \mathbf{k}_{2z}=\mathbf{k}_2\cos\theta_2=\mathbf{k}_1\sin\theta_1\coth\theta_2$

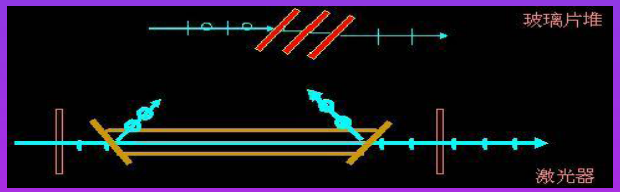
$$\mathbf{R}_{\omega\parallel}=\left\|\frac{-\cos\theta_2+\frac{\sin\theta_1}{\sin\theta_2}\cos\theta_1}{\cos\theta_2+\frac{\sin\theta_1}{\sin\theta_2}\cos\theta_1}\right\|^2=\left\|\frac{\sin 2\theta_1-\sin 2\theta_2}{\sin 2\theta_1+\sin 2\theta_2}\right\|^2$$

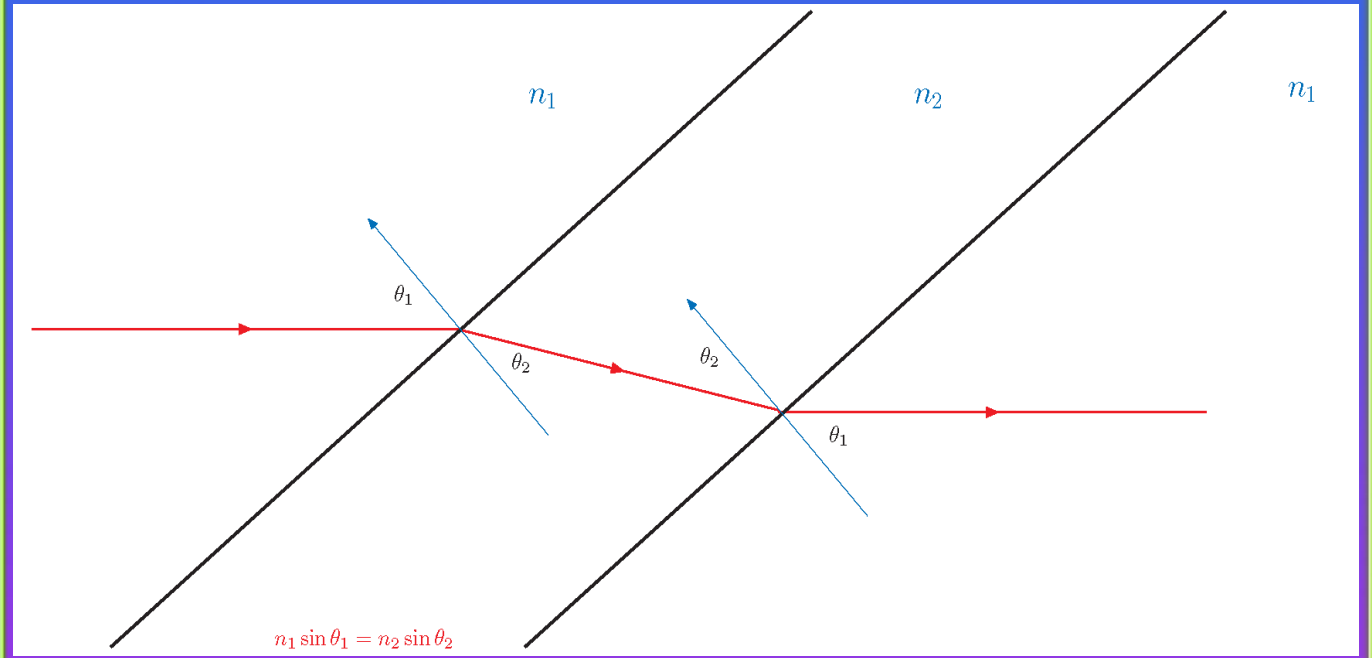
$$=\left\|\frac{\sin(\theta_1-\theta_2)\cos(\theta_1+\theta_2)}{\sin(\theta_1+\theta_2)\cos(\theta_1-\theta_2)}\right\|^2=\frac{\tan^2(\theta_1-\theta_2)}{\tan^2(\theta_1+\theta_2)}\overset{\theta_1+\theta_2=\frac{\pi}{2}}{\Longrightarrow}0$$ 布儒斯特定律

$$\mathbf{R}_{\omega\perp}=\left\|\frac{-\sin\theta_1\coth\theta_2+\cos\theta_1}{\sin\theta_1\coth\theta_2+\cos\theta_1}\right\|^2=\left[\frac{-\sin\theta_1\cos\theta_2+\cos\theta_1\sin\theta_2}{\sin\theta_1\cos\theta_2+\cos\theta_1\sin\theta_2}\right]^2$$

$$=\frac{\sin^2(\theta_1-\theta_2)}{\sin^2(\theta_1+\theta_2)}\overset{\theta_1+\theta_2=\frac{\pi}{2}}{\Longrightarrow}\cos^2 2\theta_1$$

$\mathbf{D}_{\parallel}=1-\mathbf{R}_{\parallel}$ $\qquad$ $\mathbf{D}_{\perp}=1-\mathbf{R}_{\perp}$




n_1
n_2
n_1
θ_1
θ_2
θ_2
θ_1
$n_1 \sin\theta_1 = n_2 \sin\theta_2$

电磁波在界面上的反射和折射：垂直入射和零折射率透射介质

若：$\gamma_2 = 0$, $\epsilon_2 = \mu_2 \to 0$, $\epsilon_2/\mu_2 = \epsilon_1/\mu_1$!

$$\vec{k}_1 = -\vec{k}_3 = \vec{e}_3\sqrt{\mu_1\epsilon_1}\omega \qquad \vec{k}_2 = \vec{e}_3\sqrt{\mu_2\epsilon_2}\omega \to 0$$

$$\frac{k_{2z}\mu_2}{k_2^2} = \frac{\sqrt{\epsilon_2\mu_2}\omega\mu_2}{\epsilon_2\mu_2\omega^2} = \sqrt{\frac{\mu_2}{\epsilon_2}}\frac{1}{\omega} = \frac{\mu_1}{k_1} \Rightarrow \vec{E}_{30\parallel} = \frac{\frac{k_{2z}\mu_2}{k_2^2} - \frac{\mu_1}{k_1}}{\frac{k_{2z}\mu_2}{k_2^2} + \frac{\mu_1}{k_1}}E_{10\parallel}\vec{e}_1 = 0$$

$$\frac{k_{2z}}{\mu_2} = \frac{\sqrt{\mu_2\epsilon_2}\omega}{\mu_2} = \frac{k_1}{\mu_1} \Rightarrow \vec{E}_{30\perp} = \frac{1 - \frac{\mu_1 k_{2z}}{\mu_2 k_1}}{1 + \frac{\mu_1 k_{2z}}{\mu_2 k_1}}E_{10\perp}\vec{e}_2 = 0$$

$$\vec{E}_{20\parallel} = \frac{2\frac{k_{2z}}{k_2}}{\frac{k_{2z}}{k_2} + \frac{k_2\mu_1}{k_1\mu_2}}E_{10\parallel}\vec{e}_1 = E_{10\parallel}\vec{e}_1 \qquad \vec{E}_{20\perp} = \frac{2\frac{k_{2z}}{k_2}}{1 + \frac{\mu_1 k_{2z}}{\mu_2 k_1}}E_{10\perp}\vec{e}_1 = E_{10\perp}\vec{e}_1$$

$$R_\parallel = \left\|\frac{1 - \frac{k_2^2\mu_1}{k_1 k_{2z}\mu_2}}{1 + \frac{k_2^2\mu_1}{k_1 k_{2z}\mu_2}}\right\|^2 = 0 \qquad R_\perp = \left\|\frac{1 - \frac{\mu_1 k_{2z}}{\mu_2 k_1}}{1 + \frac{\mu_1 k_{2z}}{\mu_2 k_1}}\right\|^2 = 0 \qquad D_\parallel = D_\perp = 1$$

没有反射波，完全透射！ 与全发射恰好形成对照